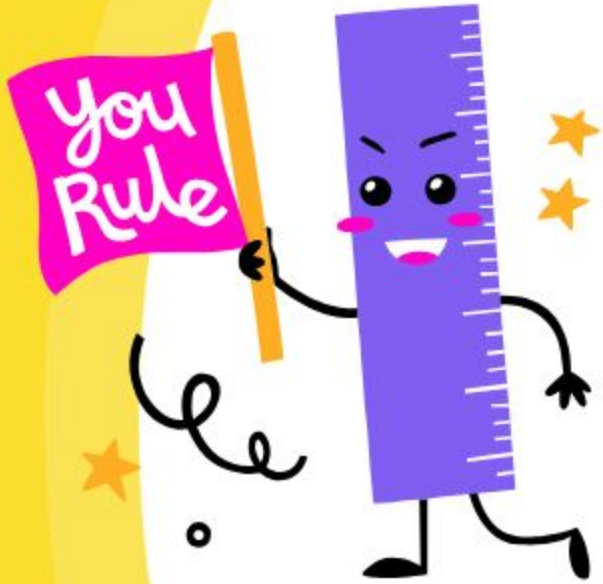
Happy International Love Data Week 2025



HARVARD LIBRARY

Whose Data is it Anyway?

More events!

Tools to Confidently Plan, Manage, and Share Research Data at Harvard

Rory Macneil (CEO) and Rob Day (Director of Sales)
Research Space (RSpace)

Becky Grady, Senior Product Manager, DMP Tool
UC Curation Center (UC3), California Digital Library

Ceilyn Boyd, Dataverse Development Project Manager
Institute for Quantitative Social Sciences (IQSS), Harvard University

HARVARD LIBRARY

Practice into Action: Tools to Confidently Plan, Manage, and Share Research Data

Open Scholarship and Research Data Services, Harvard Library

Panelists




Rory Macneil

Rob Day

CEO & Director of Sales

Research Space (RSpace)



Becky Grady

Senior Product Manager,
DMP Tool

UC Curation Center (UC3),
California Digital Library

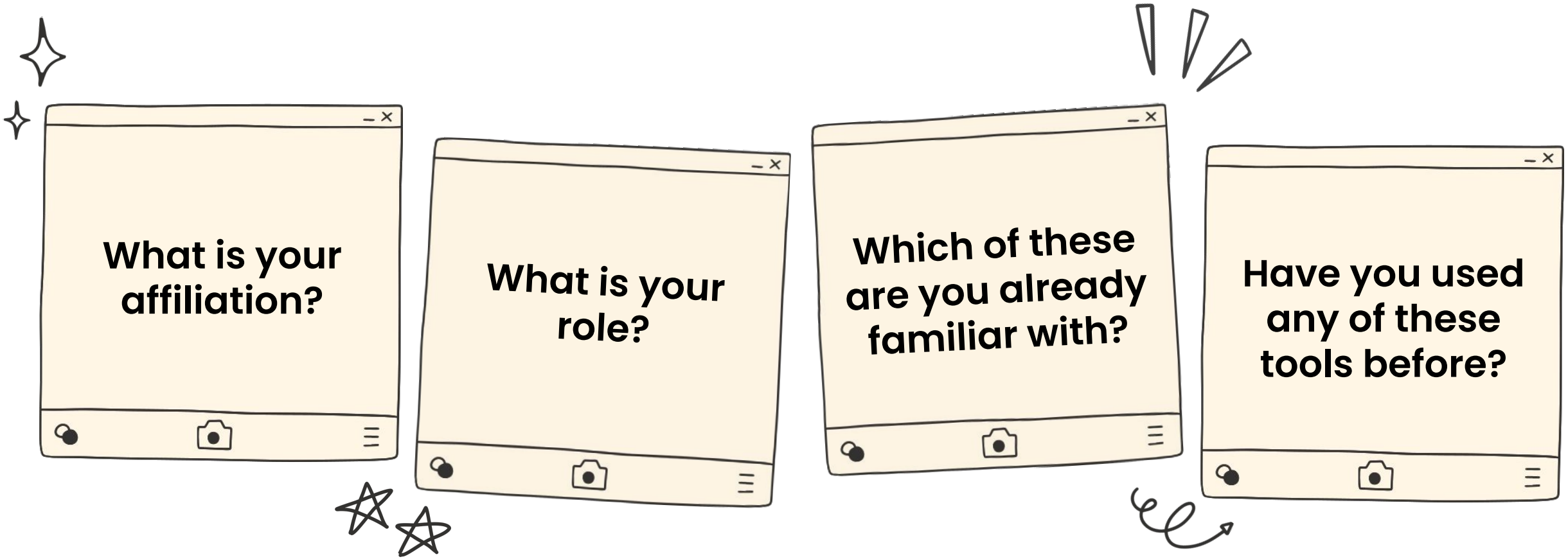


Ceilyn Boyd

Dataverse Development
Project Manager

Institute for Quantitative
Social Sciences (IQSS)

Tell us about yourself!



Webinar Outline

- **Welcome Remarks**: Understand your responsibility for the planning, management, and sharing of data at Harvard
- **Overview of RSpace**: Explore the capabilities of RSpace as an ELN and sample management system, demonstrating new features for secure and organized research data management
- **Overview of DMP Tool**: See how the DMP Tool supports creating robust machine actionable data management plans linked directly with RSpace
- **Overview of Harvard Dataverse**: See how RSpace integrates with Harvard Dataverse for streamlined data publication, with a focus on new features and services for sharing
- **Explore Integrations**: See the interoperability of RSpace, DMP Tool, and Harvard Dataverse in action!
- **Questions & Discussion**

Support for the whole Research Lifecycle



Katie Mika

Ask me about...

open scholarship

data curation

Harvard Library Open Scholarship and Research Data Services



Research Data Lifecycle by LMA RDMWG



Julie Goldman

data management

data education

Ask me about...

Harvard Library Open Scholarship and Research Data Services

Streamlining the Research Lifecycle

Ensure research products meet FAIR Data Principles

Create a Data Management Plan with DMP Tool and link to RSpace



Publish results, citing published DMP, protocols, datasets, code, ect.

Collect data and metadata in RSpace and link methods from protocols.io

Send curated collections to Harvard Dataverse repository

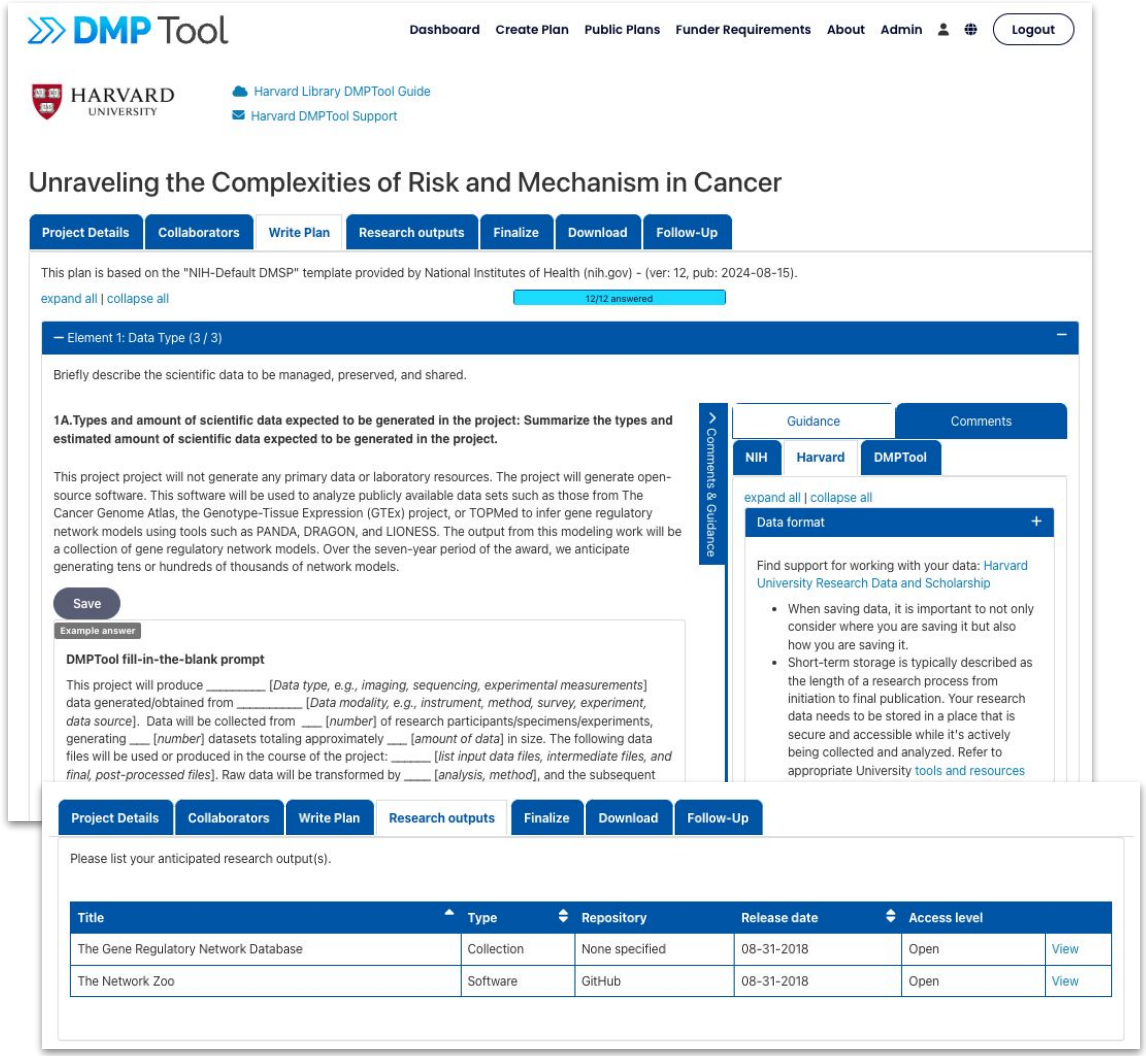
Collaborate and manage analysis workflows in RSpace

Research Data Lifecycle by LMA RDMWG

DMP Tool

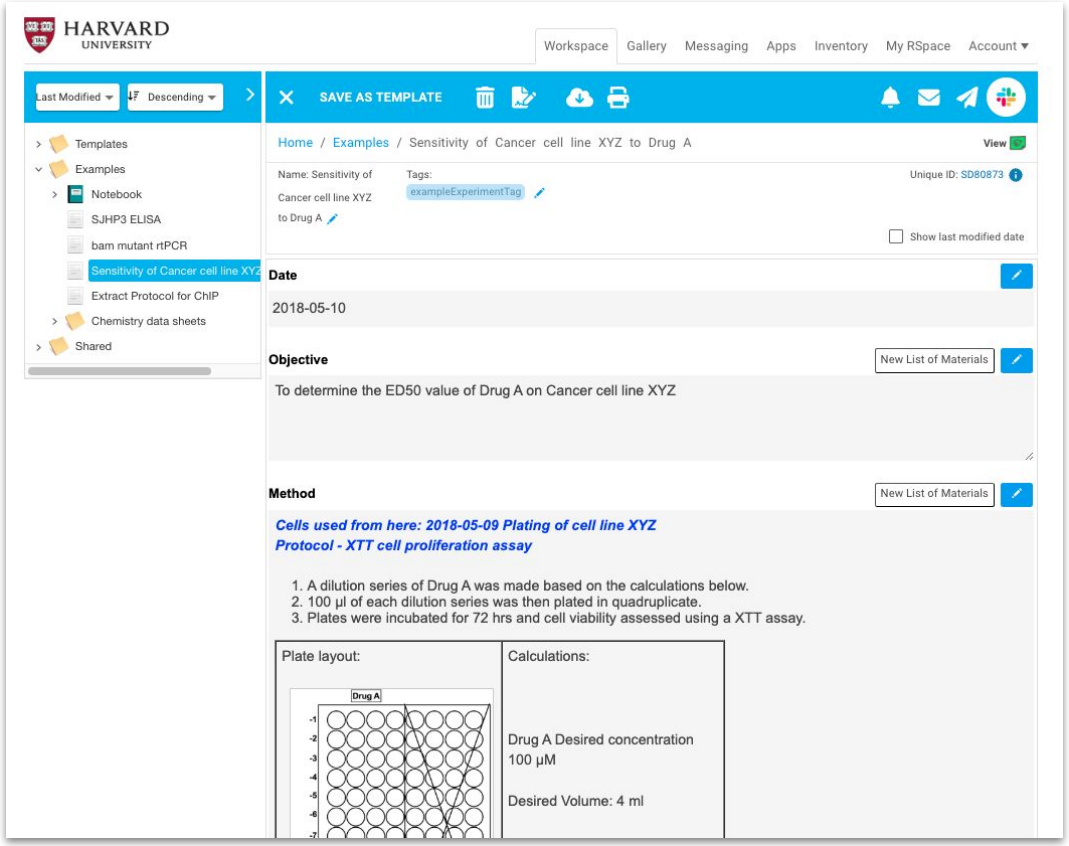
- Free tool that helps researchers create data management and sharing plans that fulfill the requirements of different funders
- DMP Tool provides templates for specific funder requirements (e.g. NIH, NSF, DOD)
- Harvard Library and partners have created guidance language and templates (general & biomedical) with example text and links to Harvard policies and resources

Harvard Library offers support for DMP Tool, workshops on writing DMS Plans, and DMS Plan consultation & review services.



RSpace

- RSpace is an open-source platform that orchestrates research workflows into FAIR data management ecosystems
- Keep working with research tools you already use, such as Protocols.io, Evernote, OneDrive, Google Drive, ChemAxon, etc.
- Link to files in file sharing apps and Harvard's research computing storage (DSL3 and below)
- Available to Principal Investigators (PI) with a Harvard appointment
  - Per user annual charge ($120 per user)
  - Free to FAS (costs are covered by the Division)

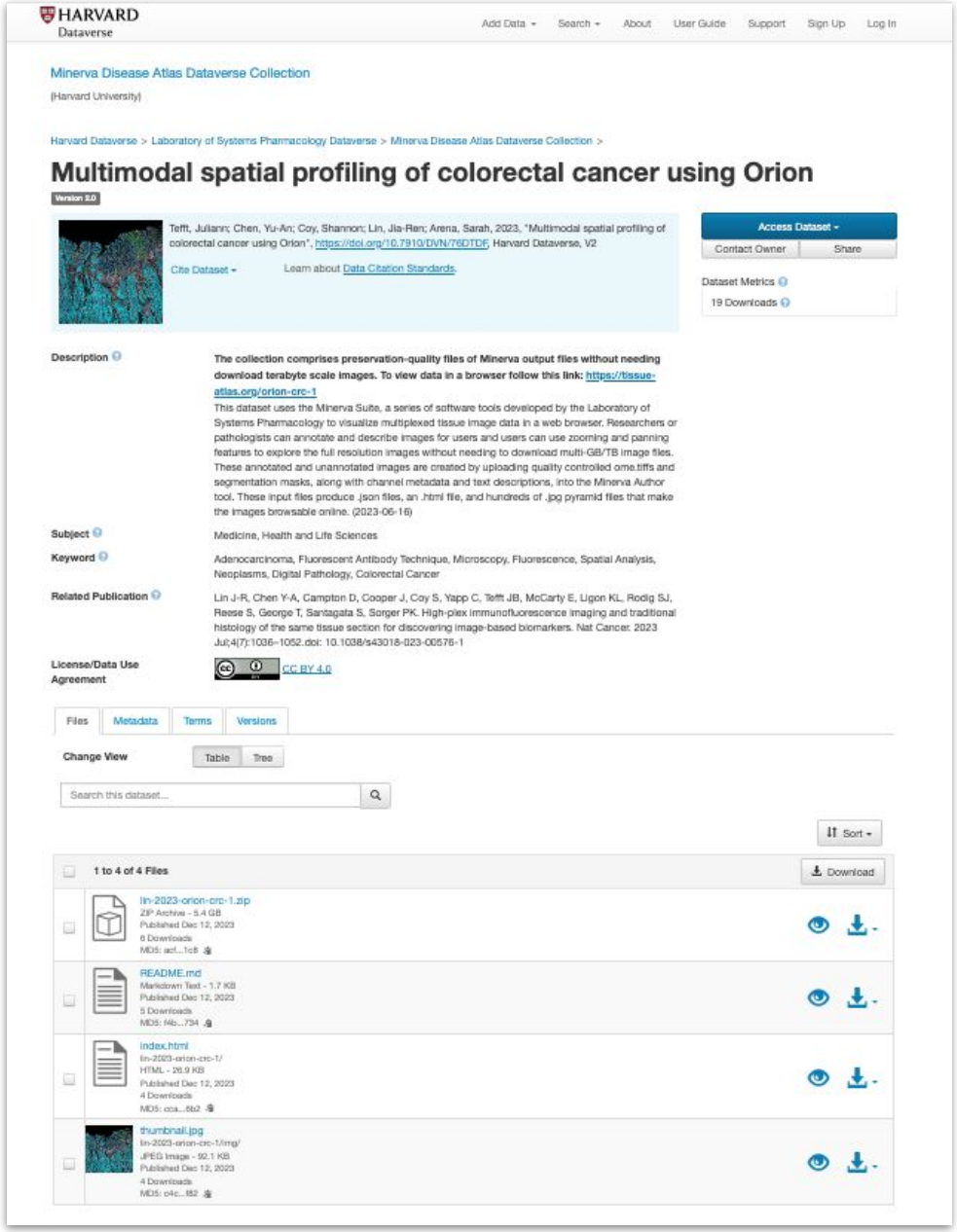


Harvard University's RSpace service is provided by Harvard University Information Technology.

Harvard Dataverse

- Free data repository open to all researchers from any discipline, both inside and outside of the Harvard community, where you can share, archive, cite, access, and explore research data
- Harvard affiliates have access to 2.5TB of free storage space, with additional options for large data
- Collaboration between Harvard Library, Harvard University Information Technology, and IQSS

Harvard Library and IQSS offer enhanced curation and deposit support for Harvard affiliates. Consult the NIH-DMSP Guidance for help writing Dataverse into a Data Management and Sharing Plan.



Confidently Plan, Manage, and Share Research Data

DMP Tool
Making best practice easier

RSpace

HARVARD
Dataverse

- Harvard University owns all research data generated through projects conducted under its authority or using its resources
- Researchers are stewards of managing and safeguarding data
  - Complying with Data Management and Sharing Plan requirements
  - Establishing clear and consistent data management expectations
  - Complying with policies and requirements for data sharing
- Strongly recommend utilizing tools supported by the institution to meet planning, management, and sharing goals



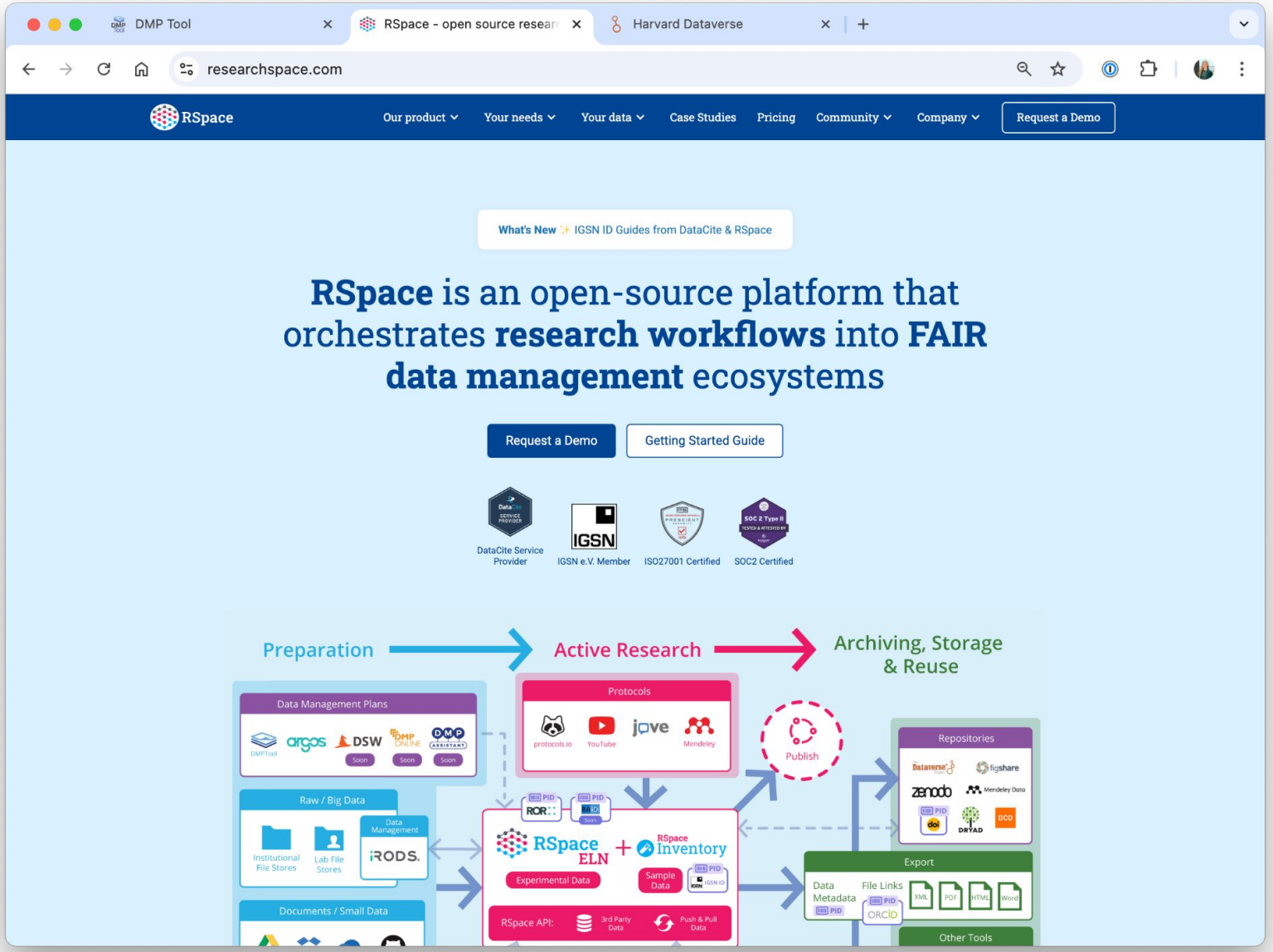


Rory Macneil

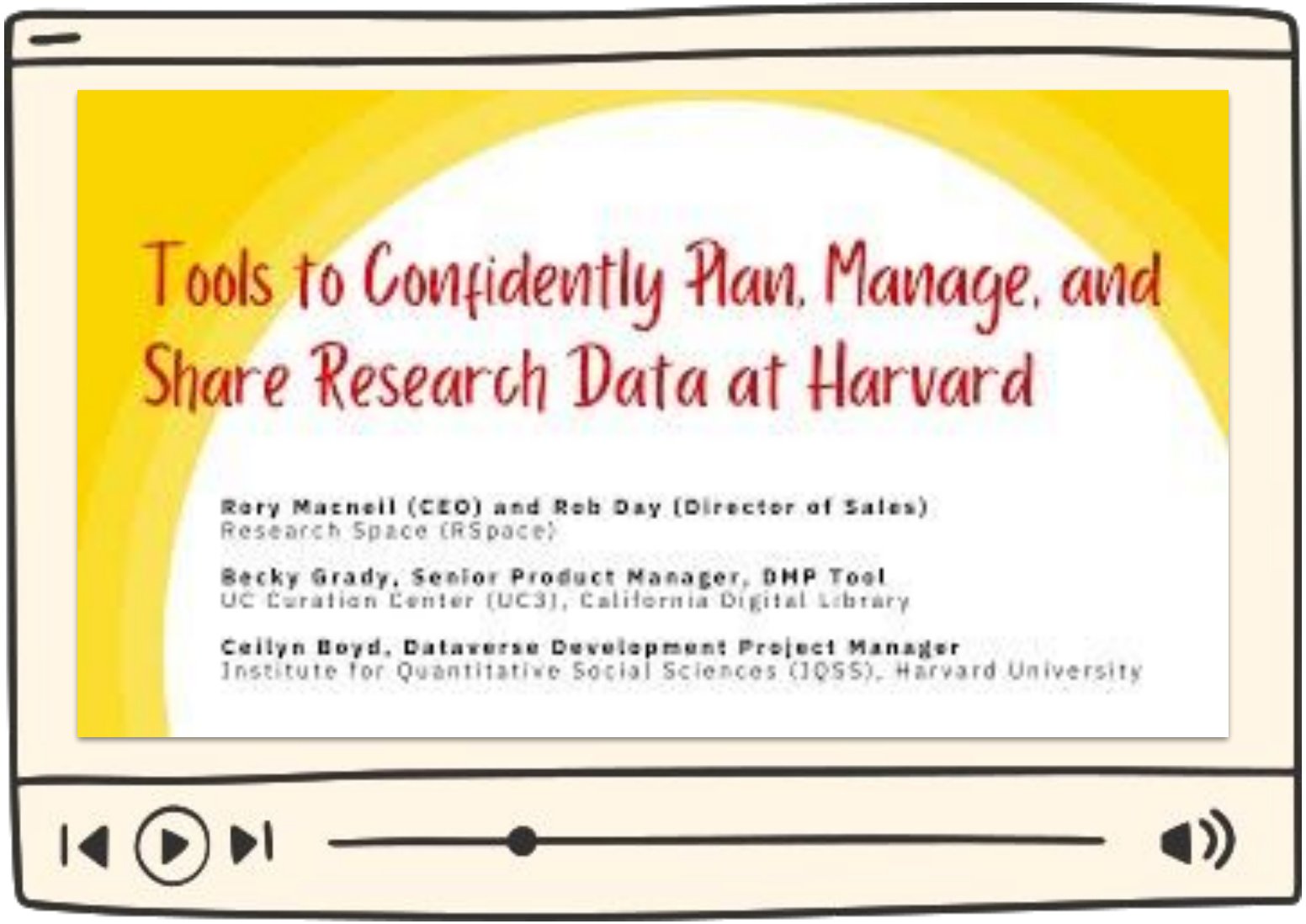
Rob Day

CEO & Director of Sales

Research Space (RSpace)



Find RSpace Demo in the YouTube Recording

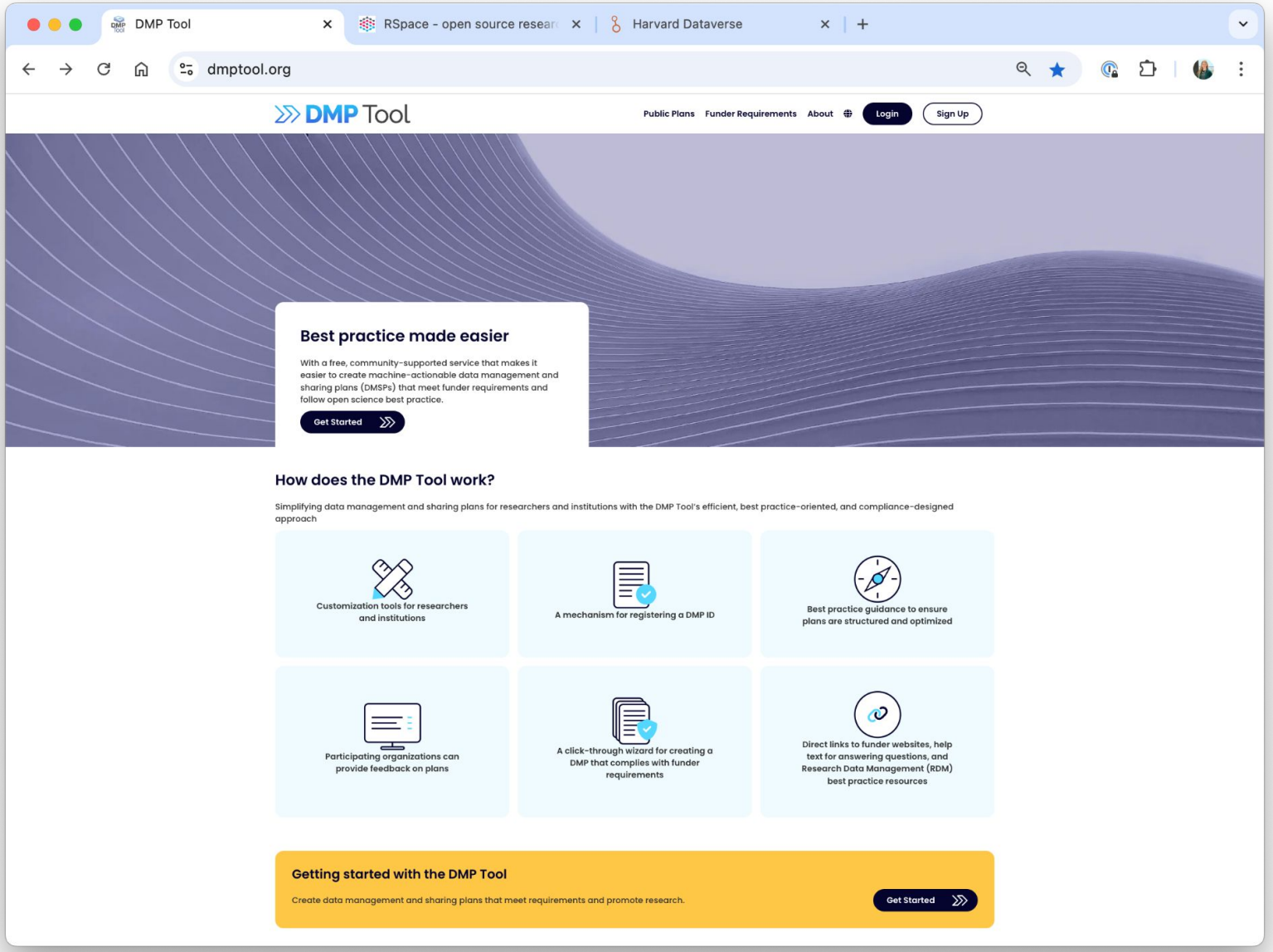




Becky Grady

Senior Product Manager,
DMP Tool

UC Curation Center (UC3),
California Digital Library



California Digital Library (CDL)/UC3



CDL founded by the University of California in 1996

Part of the UC Office of the President

University of California Curation Center (UC3) is CDL's program concerned with maintaining, preserving, and adding value to digital research data

- Research data management
- Data publication and data metrics
- Persistent identifiers
- Digital preservation
- Data/software skills training

Our current tool

DMP Tool

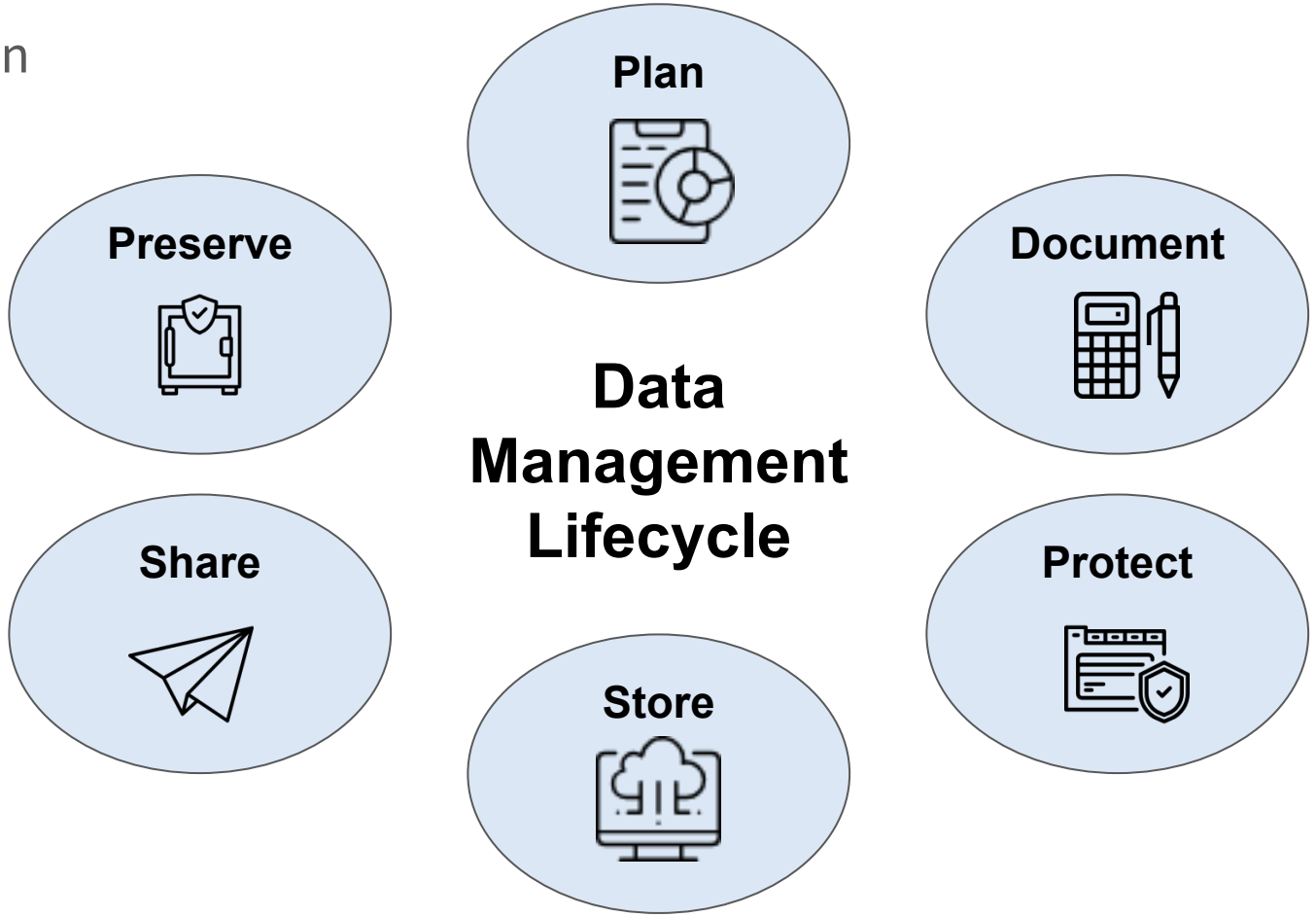
Making best practice easier

What is a DMP (also known as DMSP)?

Data **M**anagement (& **S**haring) **P**lan

A document that addresses how you will **manage** and **secure** your data throughout the lifecycle of a research project.

Can be both a **required** document for grants and a **living** document for research planning purposes



DMP Tool helps create Data Management Plans



- Is completely free and open-source
- Includes easy-to-use walk-through wizard
- Uses funder- and field- specific templates
- Allows organization-specific customization
- Offers best practice guidance to ensure DMPs are structured and optimized
- Registers DMP-IDs (DOIs) for persistent identification

Responsive to funder requirements

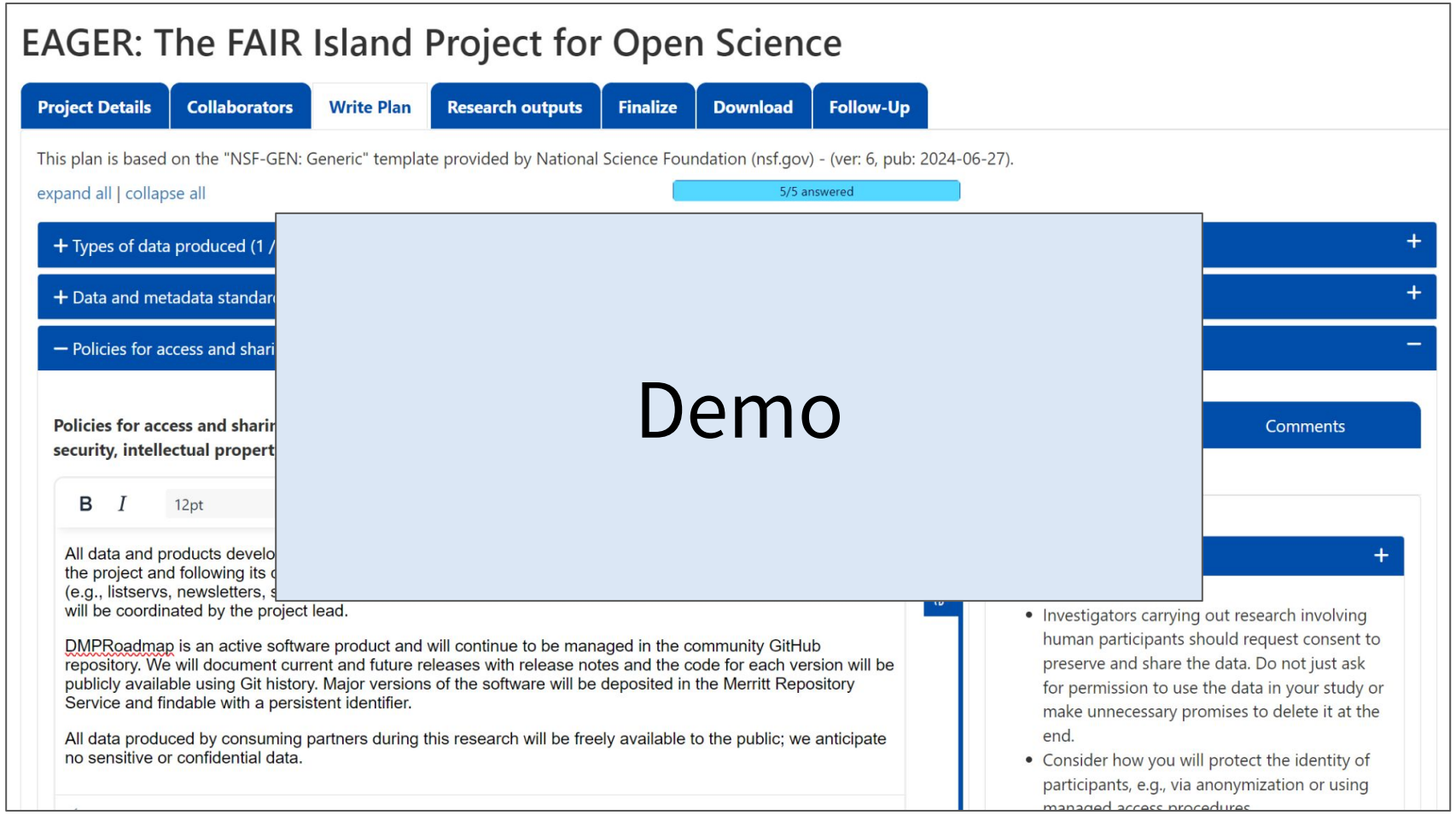
- Addressing research needs after NSF and NIH started requirement plans for managing research data
- Bringing together expertise and reduce cost across universities by centralizing data management planning needs
- Making it easy to meet requirements and best practices
- Evolving over time in response to updated policies and funding requirements



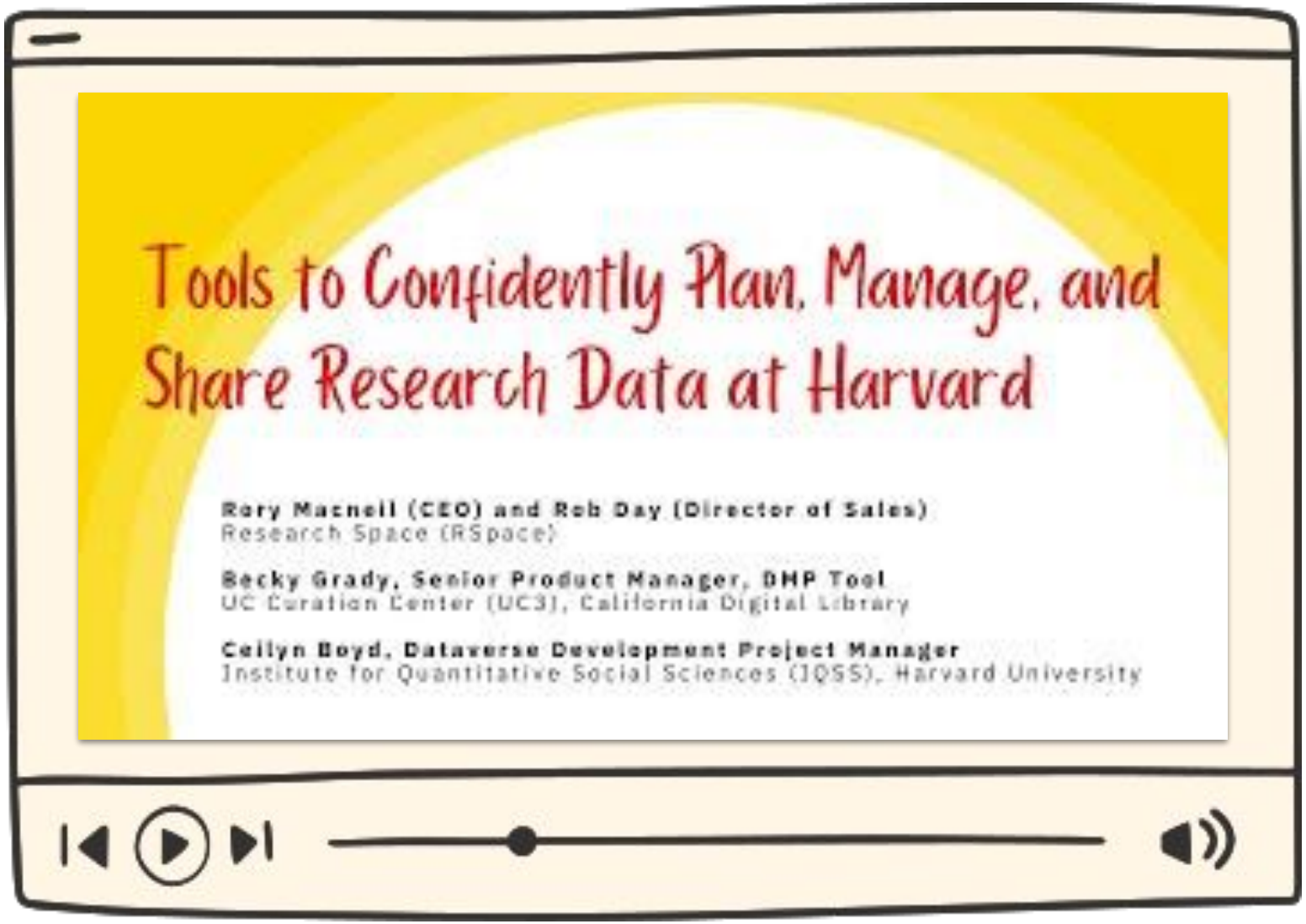
(b) DATA MANAGEMENT PLANS.—

(1) IN GENERAL.—The Director shall require that every proposal for funding for research include a machine-readable data management plan that includes a description of how the awardee will archive and preserve public access to data, software, and code developed as part of the proposed project.

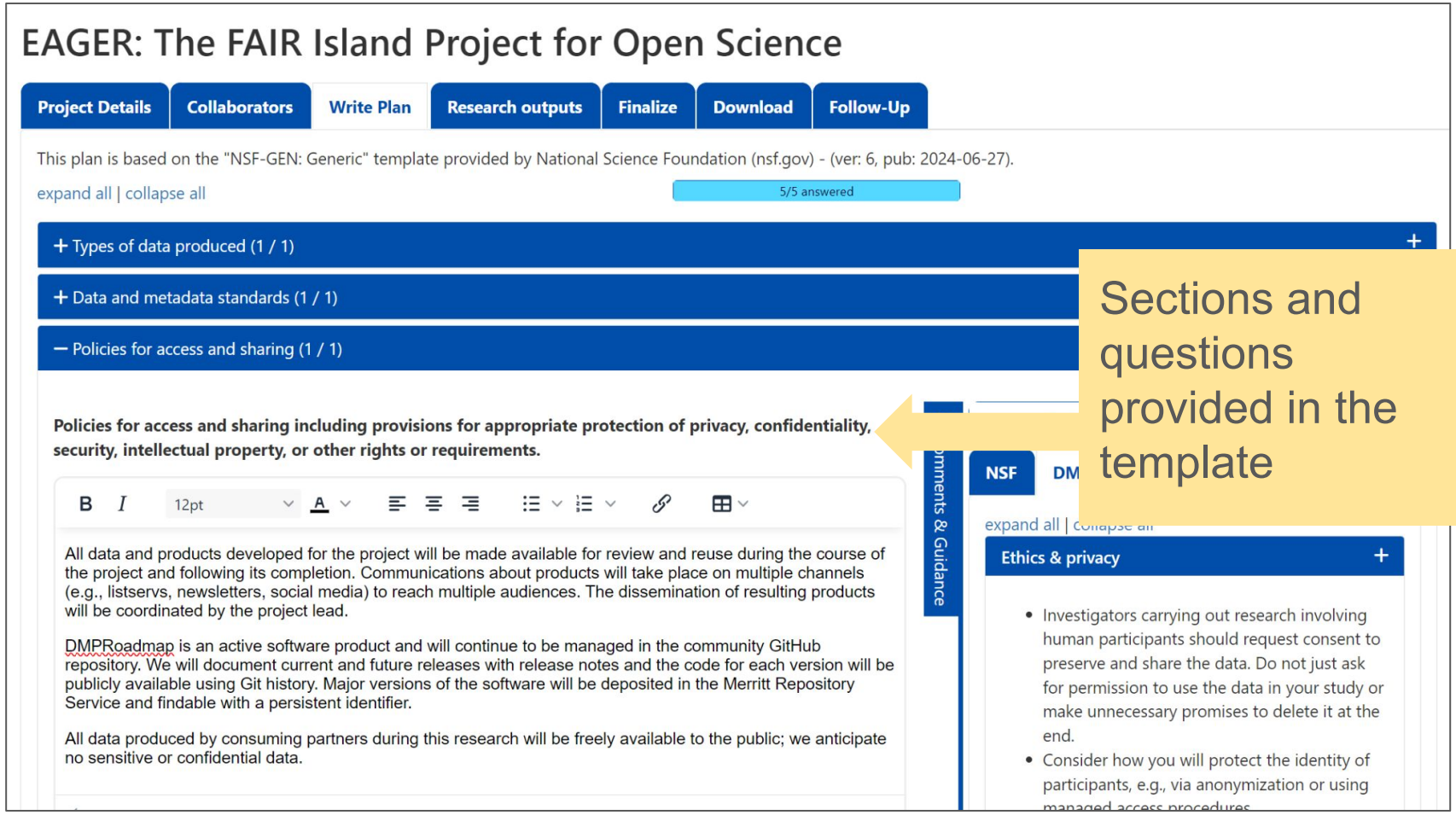
Researchers write plans using funder templates



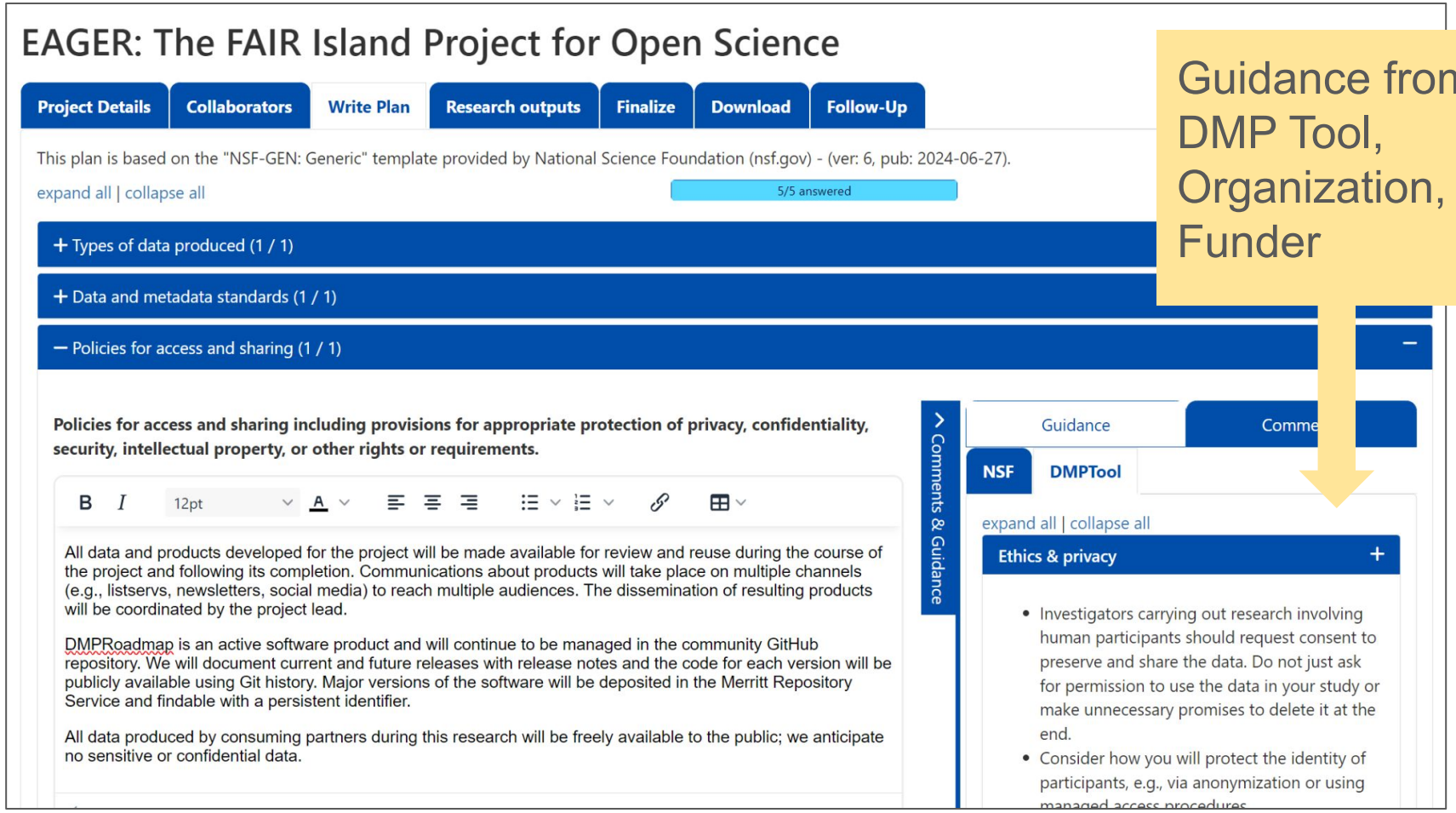
Find DMP Tool Demo in the YouTube Recording



Researchers write plans using funder templates



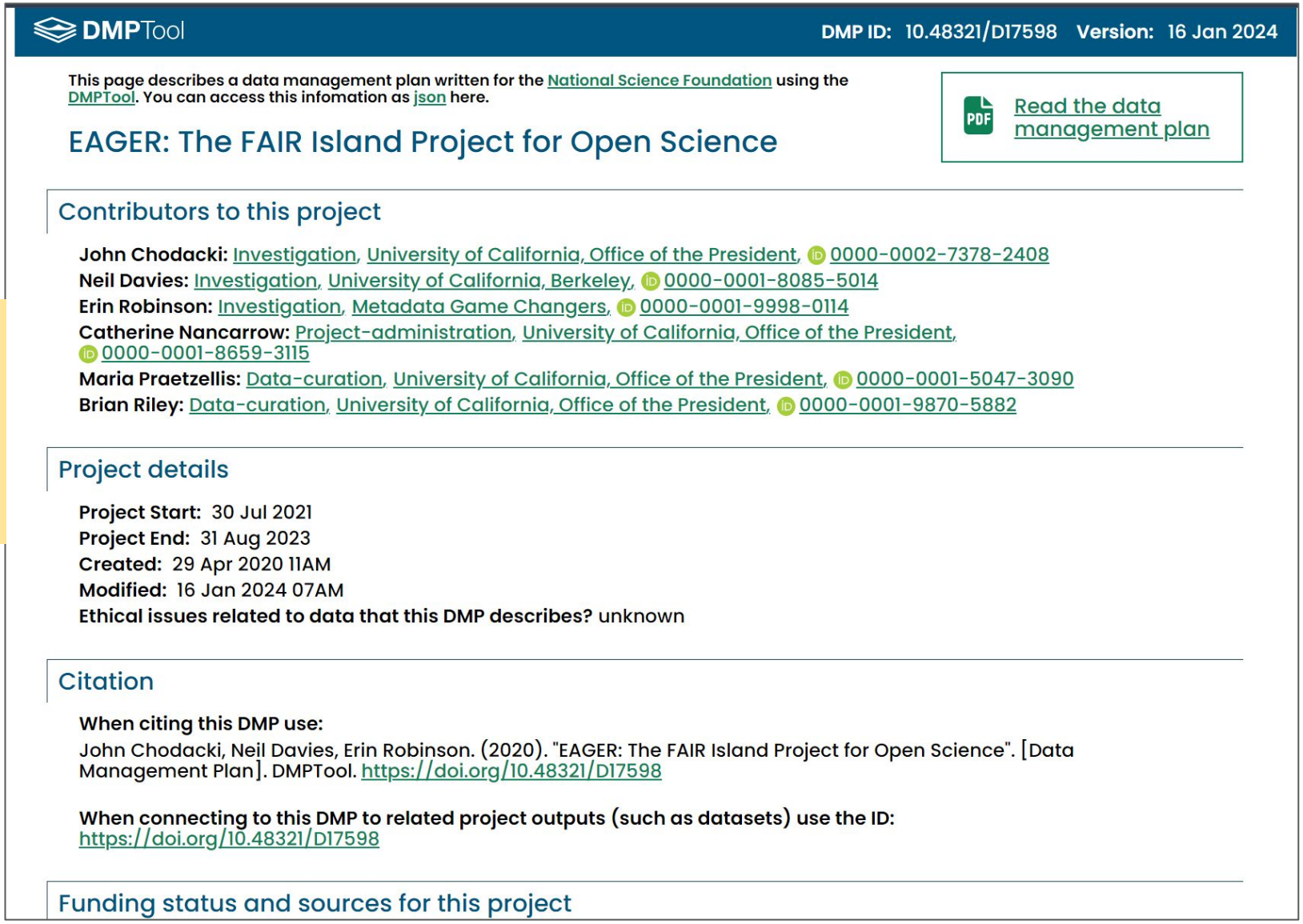
Researchers write plans using funder templates



Guidance from DMP Tool, Organization, and Funder

Sharable landing page if DMP is registered

Machine-readable metadata, persistent identifier, links to outputs and plan



DMPTool DMP ID: 10.48321/D17598 Version: 16 Jan 2024

This page describes a data management plan written for the National Science Foundation using the DMPTool. You can access this infomation as json here.

Read the data management plan

EAGER: The FAIR Island Project for Open Science

Contributors to this project

John Chodacki: Investigation, University of California, Office of the President, 0000-0002-7378-2408
Neil Davies: Investigation, University of California, Berkeley, 0000-0001-8085-5014
Erin Robinson: Investigation, Metadata Game Changers, 0000-0001-9998-0114
Catherine Nancarrow: Project-administration, University of California, Office of the President, 0000-0001-8659-3115
Maria Praetzellis: Data-curation, University of California, Office of the President, 0000-0001-5047-3090
Brian Riley: Data-curation, University of California, Office of the President, 0000-0001-9870-5882

Project details

Project Start: 30 Jul 2021
Project End: 31 Aug 2023
Created: 29 Apr 2020 11AM
Modified: 16 Jan 2024 07AM
Ethical issues related to data that this DMP describes? unknown

Citation

When citing this DMP use:
John Chodacki, Neil Davies, Erin Robinson. (2020). "EAGER: The FAIR Island Project for Open Science". [Data Management Plan]. DMPTool. https://doi.org/10.48321/D17598

When connecting to this DMP to related project outputs (such as datasets) use the ID:
https://doi.org/10.48321/D17598

Funding status and sources for this project

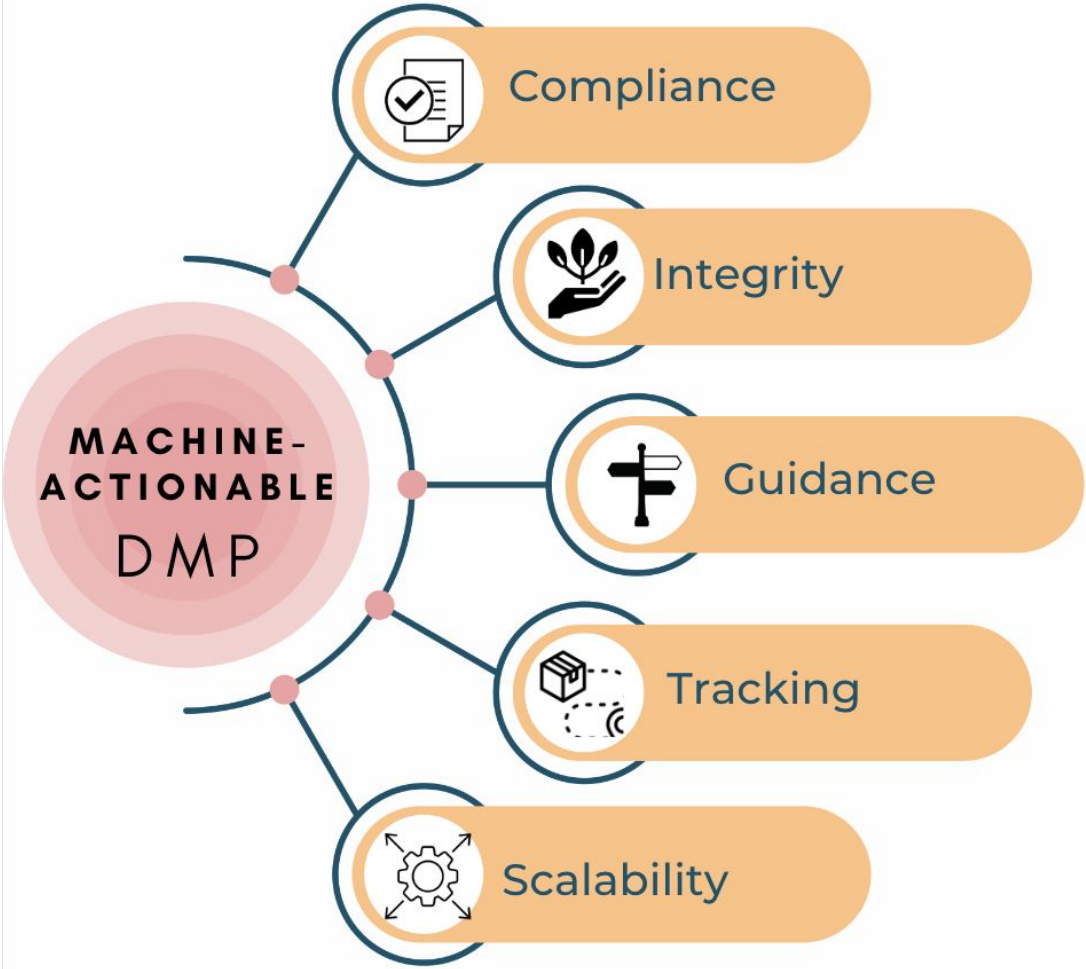
Rebuilding the tool to offer machine-actionable functionality and improve user experience



- Outdated technology in current site is difficult to update
- New data model for templates, questions
- Rethink model of DMPs in response to feedback
- Read more here: blog post about DMP Tool rebuild
- Major changes:
  - Project-level to contain multiple DMPs
  - More flexibility in creating templates with PIDs, structured questions
  - Designing around a living document
  - Much greater API functionality

What is a maDMP?

- **Structured**, machine-readable versions of traditional narrative based DMPs
- Enables **automation** & **integration** with administrative & research workflows
- Living, versioned documents that are **updatable** over time



Goal is to make entire DMP machine-actionable

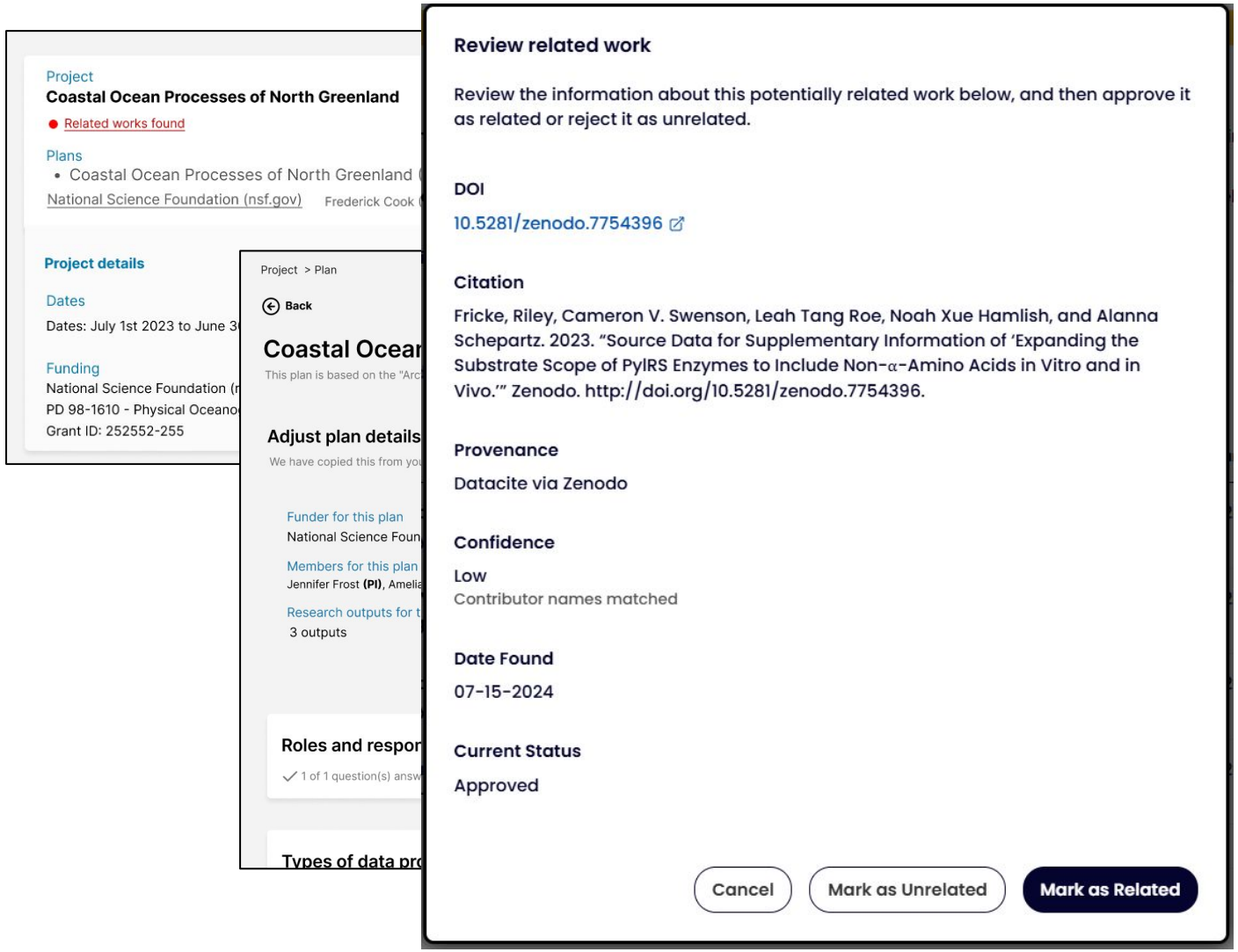
Eventual goal is to:

- Allow all functionality within the DMP Tool to be done through the API, not website
- Support integrations and interoperability with other relevant DMP and research system systems with full plan in JSON, API or upload
- Offer more structured question types beyond text boxes

API Documentation

https://github.com/CDLUC3/dmsp_api_prototype/wiki/API-Overview

Making DMPs active documents, easier to use



Plan Changes

- Project level to contain multiple DMPs
- Use templates other than from selected funder
- Upload DMP created elsewhere
- Change project/plan details after creation
- Mark complete without publishing publicly
- Find related works

DMP Tool

Making best practice easier

Thank you!

Reach out with questions, feedback: becky.grady@ucop.edu

DMP Tool Helpdesk: dmptool@ucop.edu

Follow DMP Tool updates: https://blog.dmptool.org/

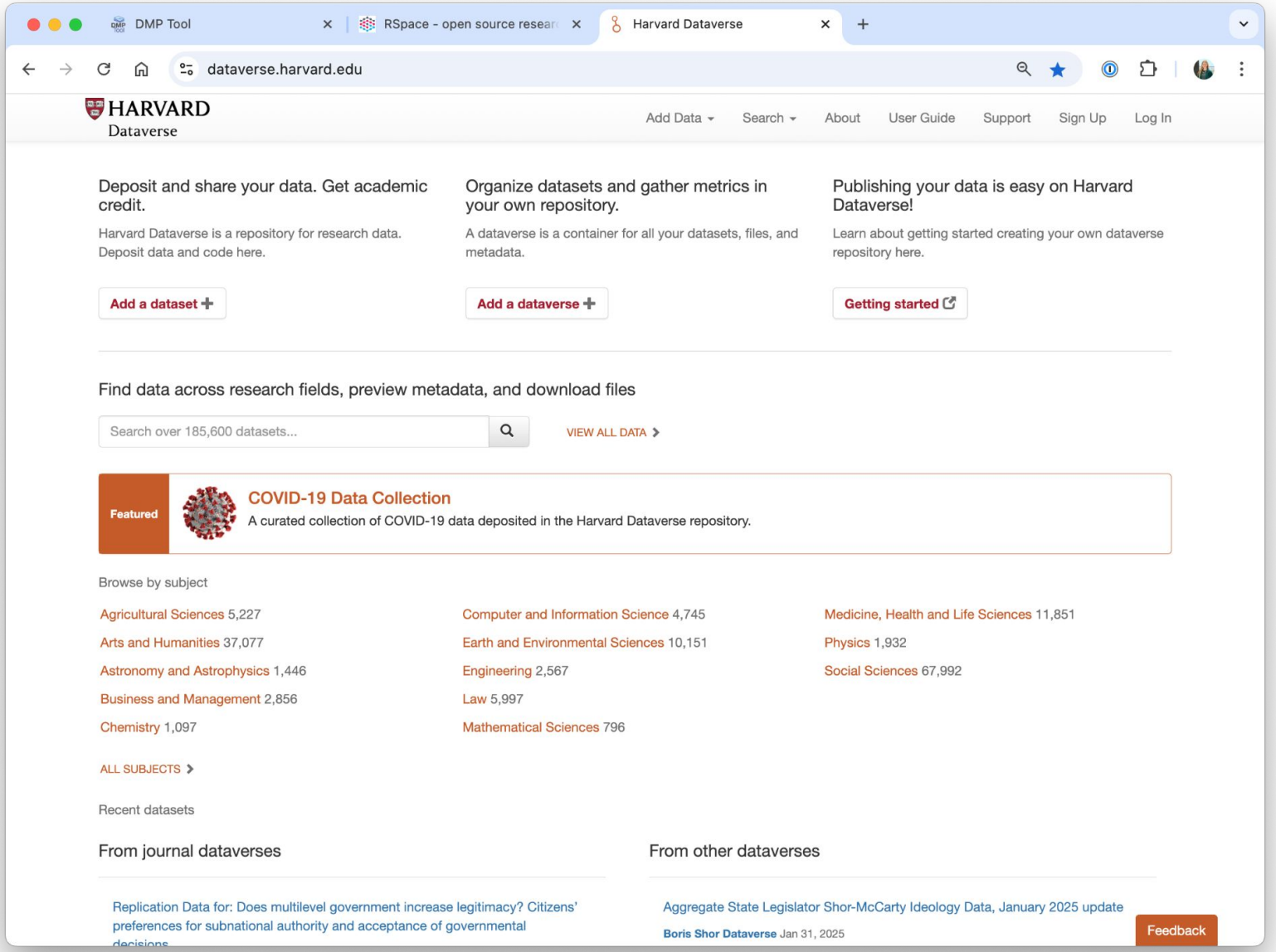
Learn about our Machine-Actionable Plan Pilot Project: https://bit.ly/mappilot



Ceilyn Boyd

Dataverse Development Project Manager

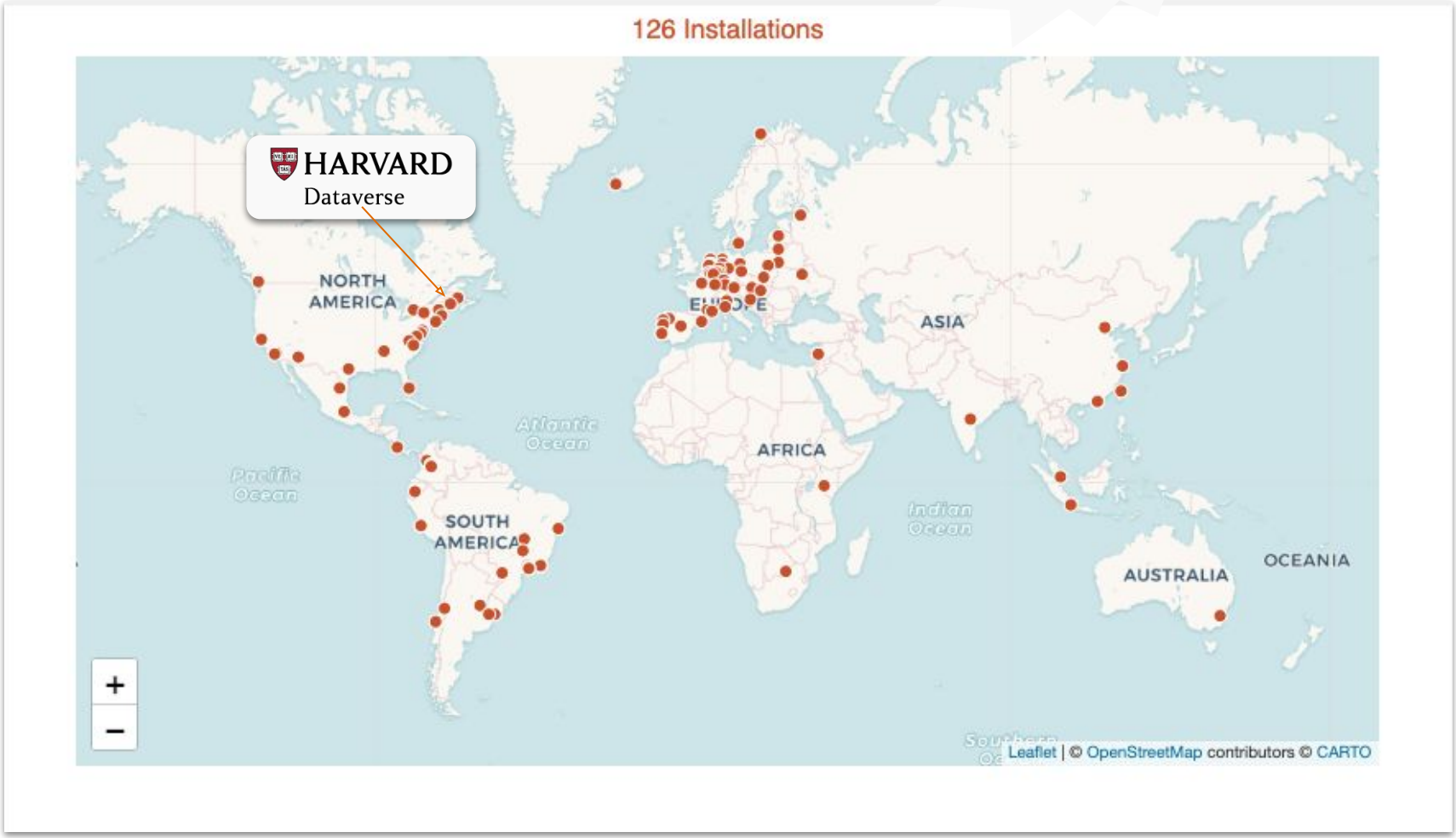
Institute for Quantitative Social Sciences (IQSS)



Dataverse Project

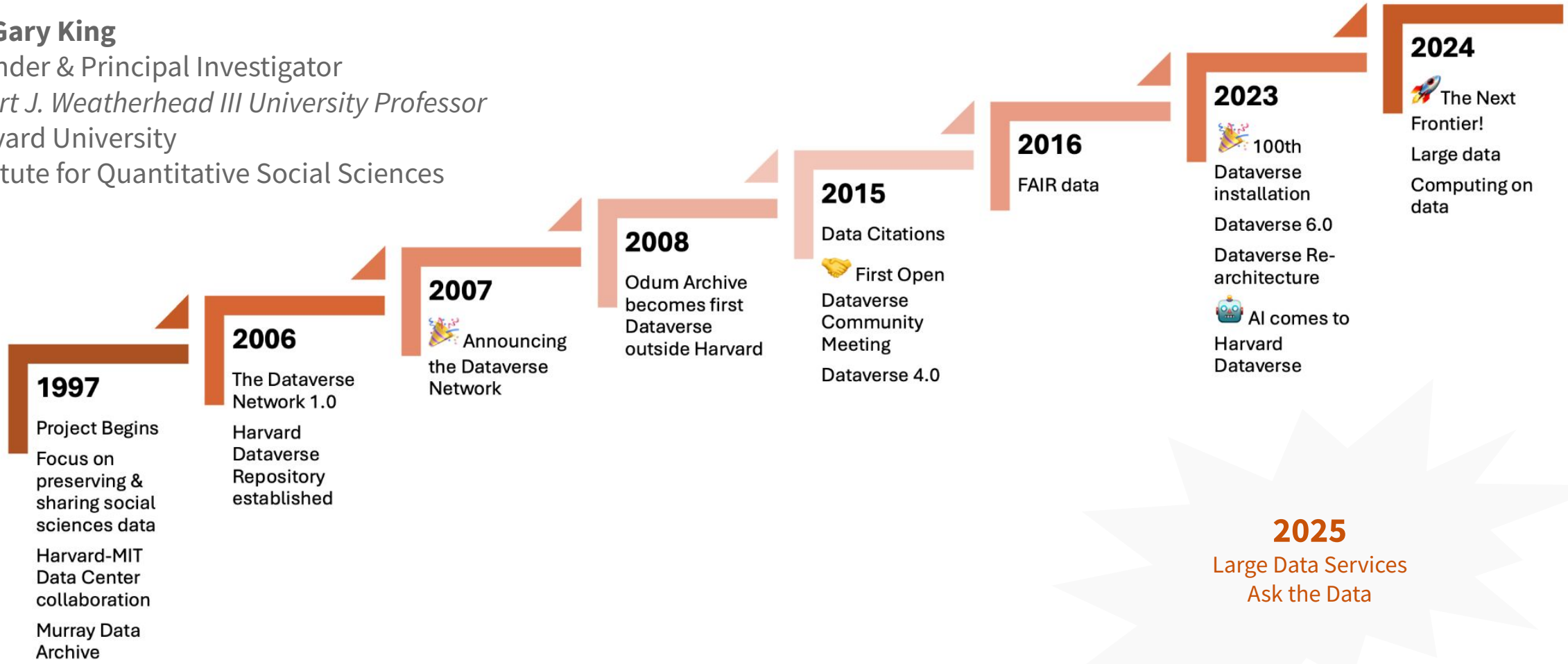
38 Countries

- Open Source
- Data Repository Software Platform
- 126 Installations
- Active Community
- FAIR support
- Standards support
- Many features
- Previewers
- External tools
- Integrations



Dataverse Project Timeline

Dr. Gary King
Founder & Principal Investigator
Albert J. Weatherhead III University Professor
Harvard University
Institute for Quantitative Social Sciences



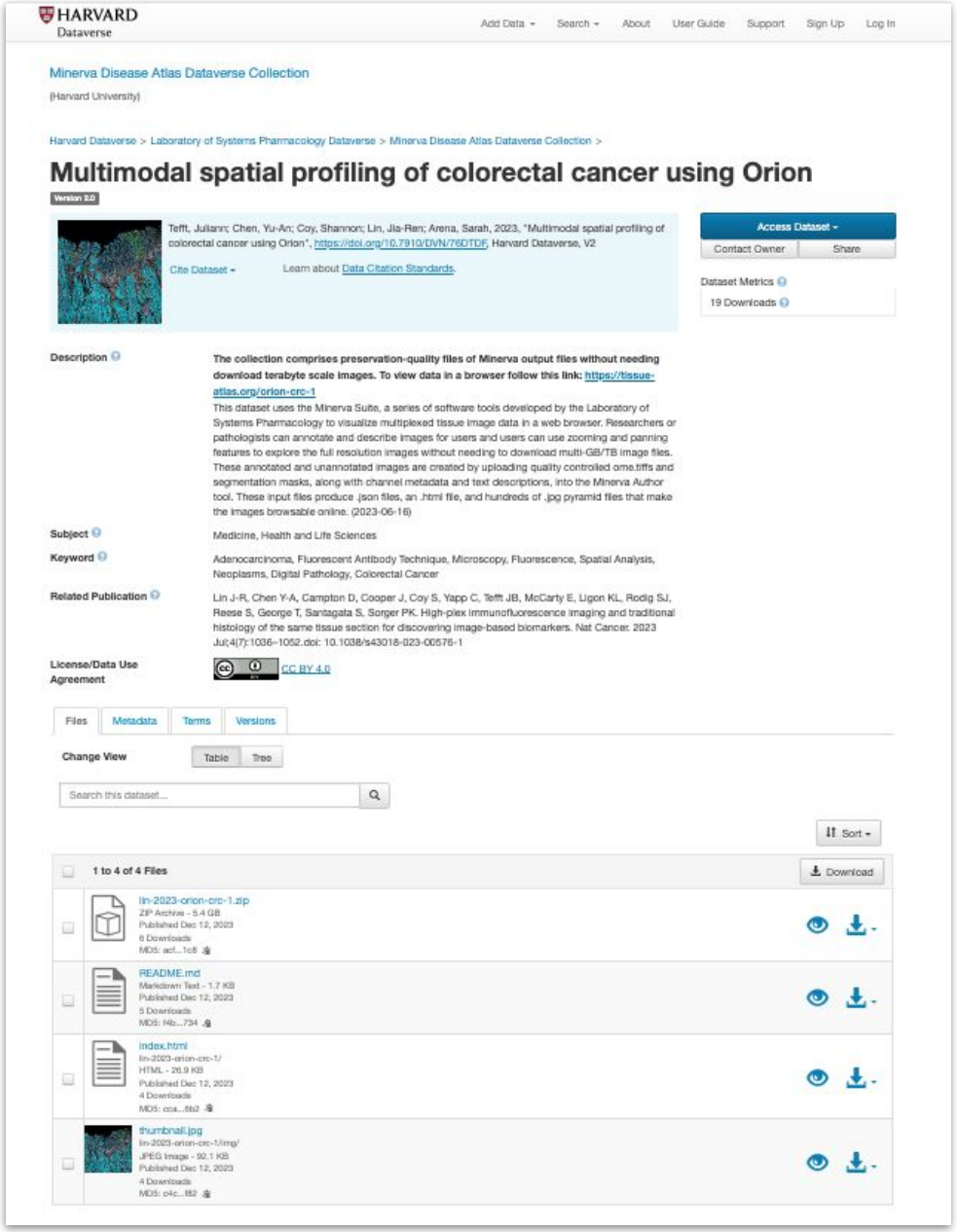
Dataverse Repository Platform Features

- Support for FAIR Data Principles
- Data citation for datasets and files
- OAI-PMH (Harvesting)
- APIs for interoperability and custom integrations
- API client libraries
- DataCite integration
- Login via Shibboleth
- Login via ORCID, Google, GitHub, or Microsoft
- Login via OpenID Connect (OIDC)
- Internationalization
- Versioning
- Restricted files
- Embargo
- Custom licenses
- Custom terms of use
- Publishing workflow support
- File hierarchy
- File previews
- Preview and analysis of tabular files
- Usage statistics and metrics
- Guestbook
- Fixity checks for files
- File download in R and TSV format
- Faceted search
- Customization of collections
- Private URL
- Widgets
- Notifications
- Schema.org JSON-LD
- External tools
- External vocabulary
- Dropbox integration
- GitHub integration
- Integration with Jupyter notebooks
- User management
- Curation status labels
- Branding
- Backend storage on S3 or Swift
- Direct upload and download for S3
- Export data in BagIt format
- Post-publication automation (workflows)
- Pull header metadata from Astronomy (FITS) files
- Provenance
- Support for rsync
- Auxiliary files for data files

https://dataverse.org/software-features

Harvard Dataverse

- Free data repository open to all researchers from any discipline, both inside and outside of the Harvard community, where you can share, archive, cite, access, and explore research data
- Collaboration between Harvard Library, Harvard University Information Technology, and IQSS
- All file types and disciplines are welcome
- We do not currently support sensitive data. All data must be de-identified.



Harvard Dataverse storage

Harvard Users	Other Users	Large Data
2.5T per account free	1T per account free	Fee-based tape & disk options

Maximum File Size

Free Storage Options	*Large Data*
Self-deposit: 2.5GB per file Contact us for larger files	Varies according to storage option

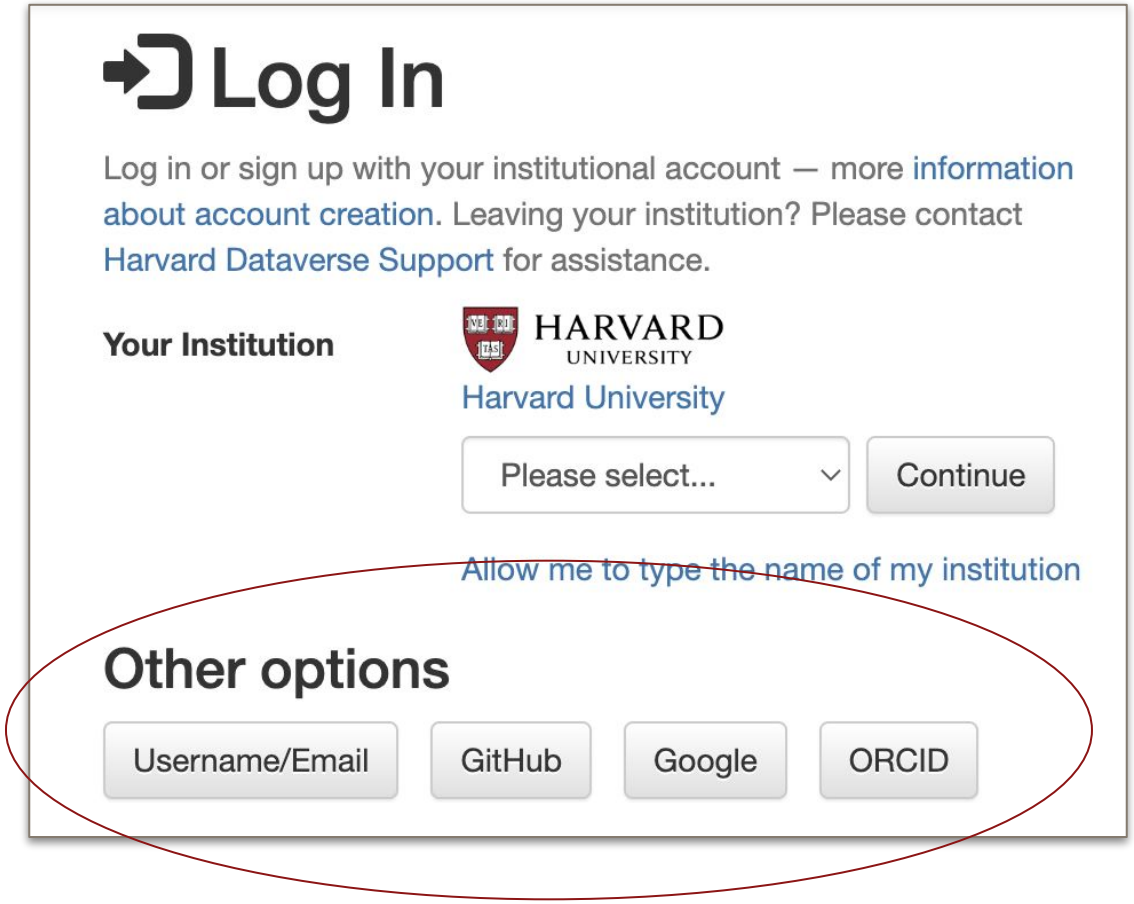
Get started

dataverse.harvard.edu

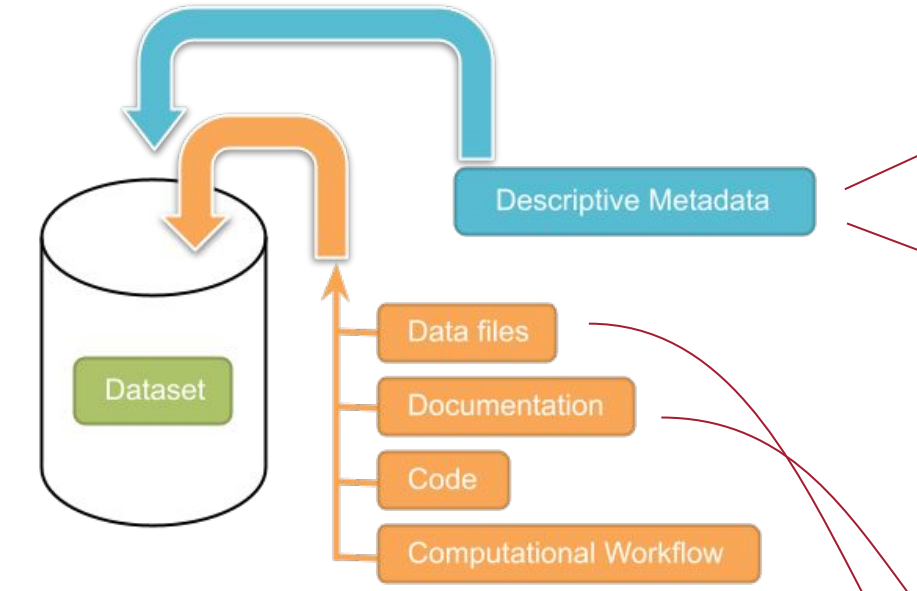
Create account	Create dataset	Publish dataset	Create dataset versions*
Harvard Key	Upload files	Receive data citation	Add more metadata and files to your dataset while your project continues
GitHub	Add metadata	Harvard Dataverse automatically shares metadata with search engines & other repositories	
ORCID	Add licenses		
Gmail	Link to other publications		

Create an account

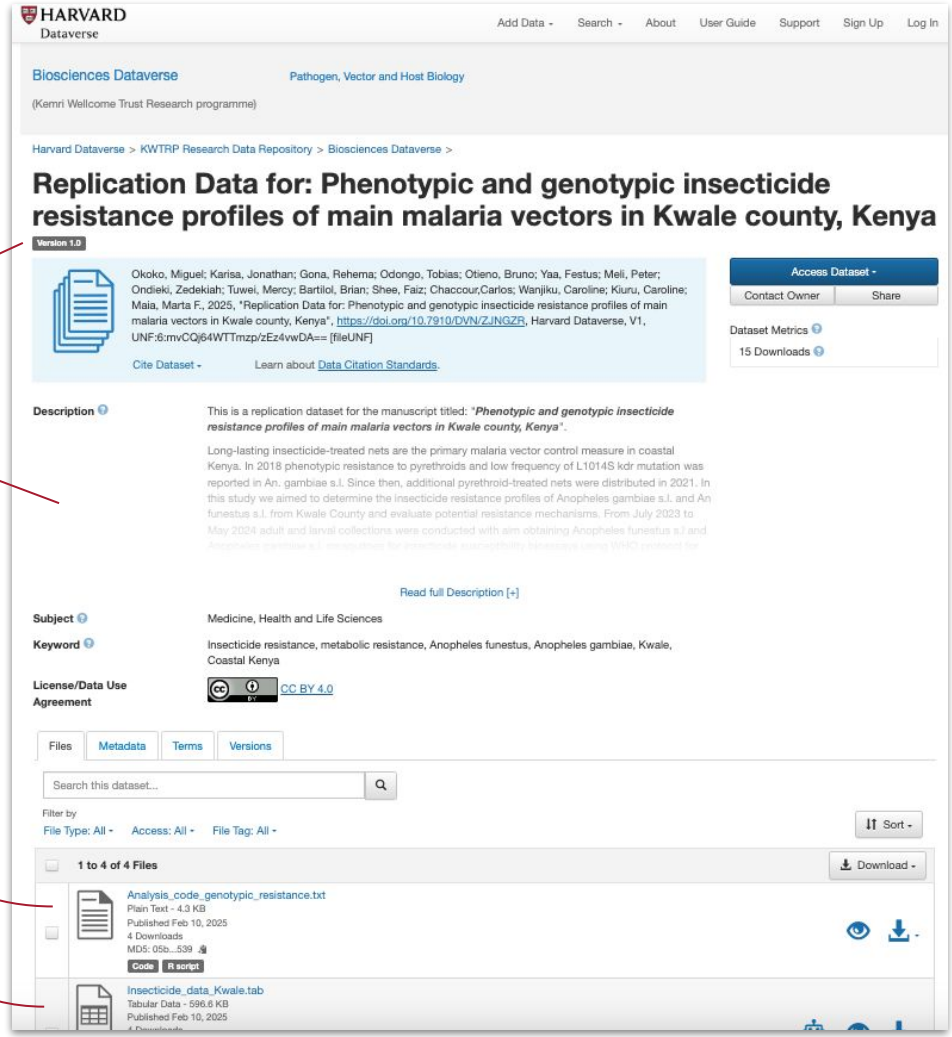
Harvard Key +



Create a dataset, upload files & folders, edit your draft

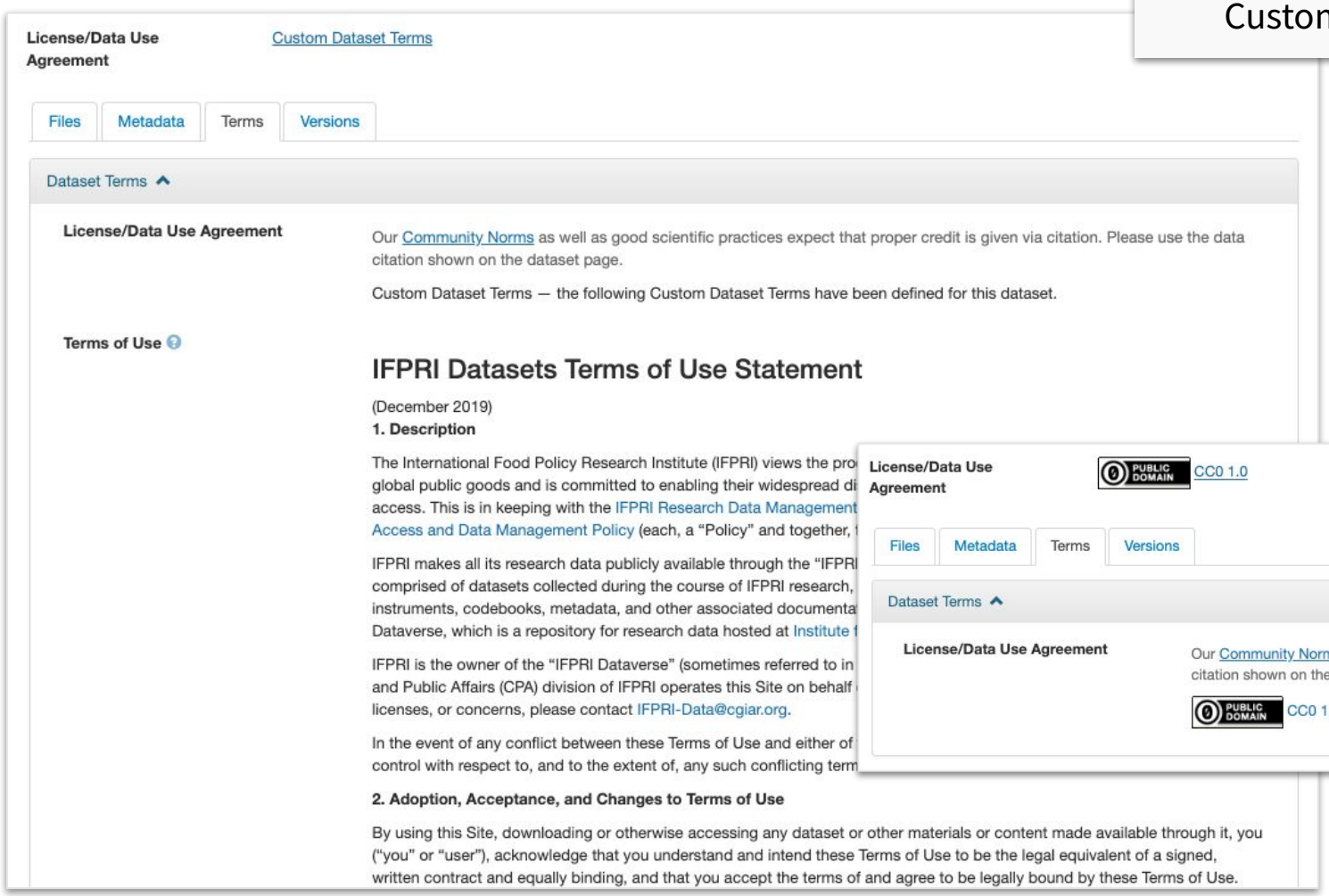


Container for your data, documentation, code, and computational workflow.

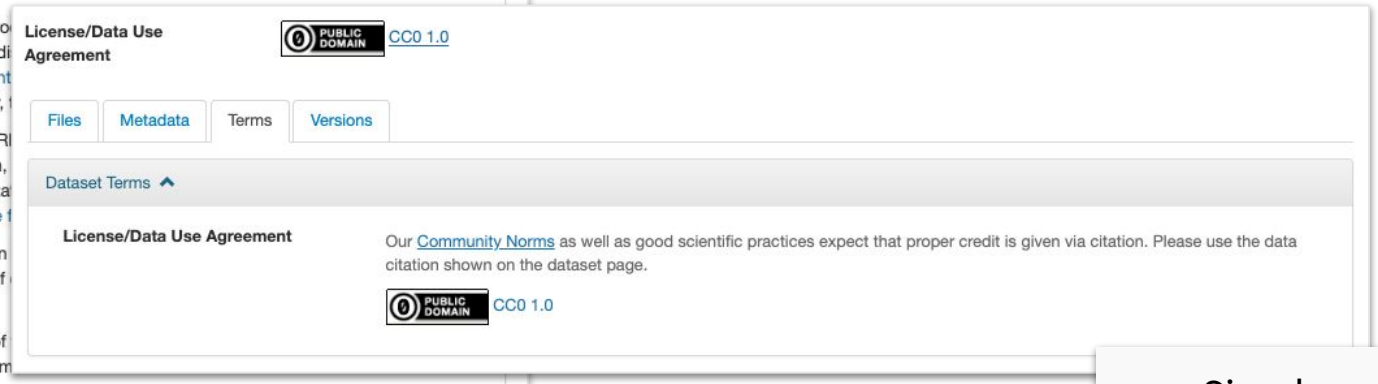


Apply license/s, set terms of use

Custom



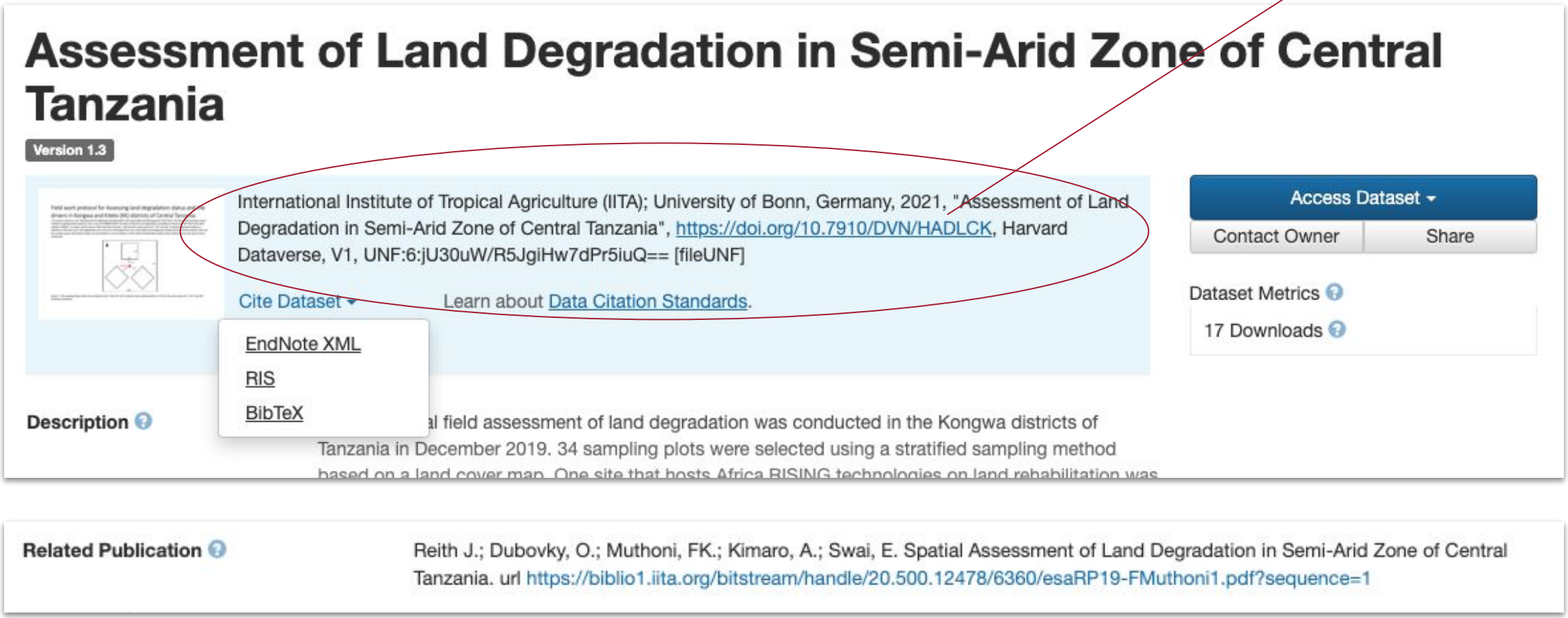
Choose simple or custom terms of use



Simple

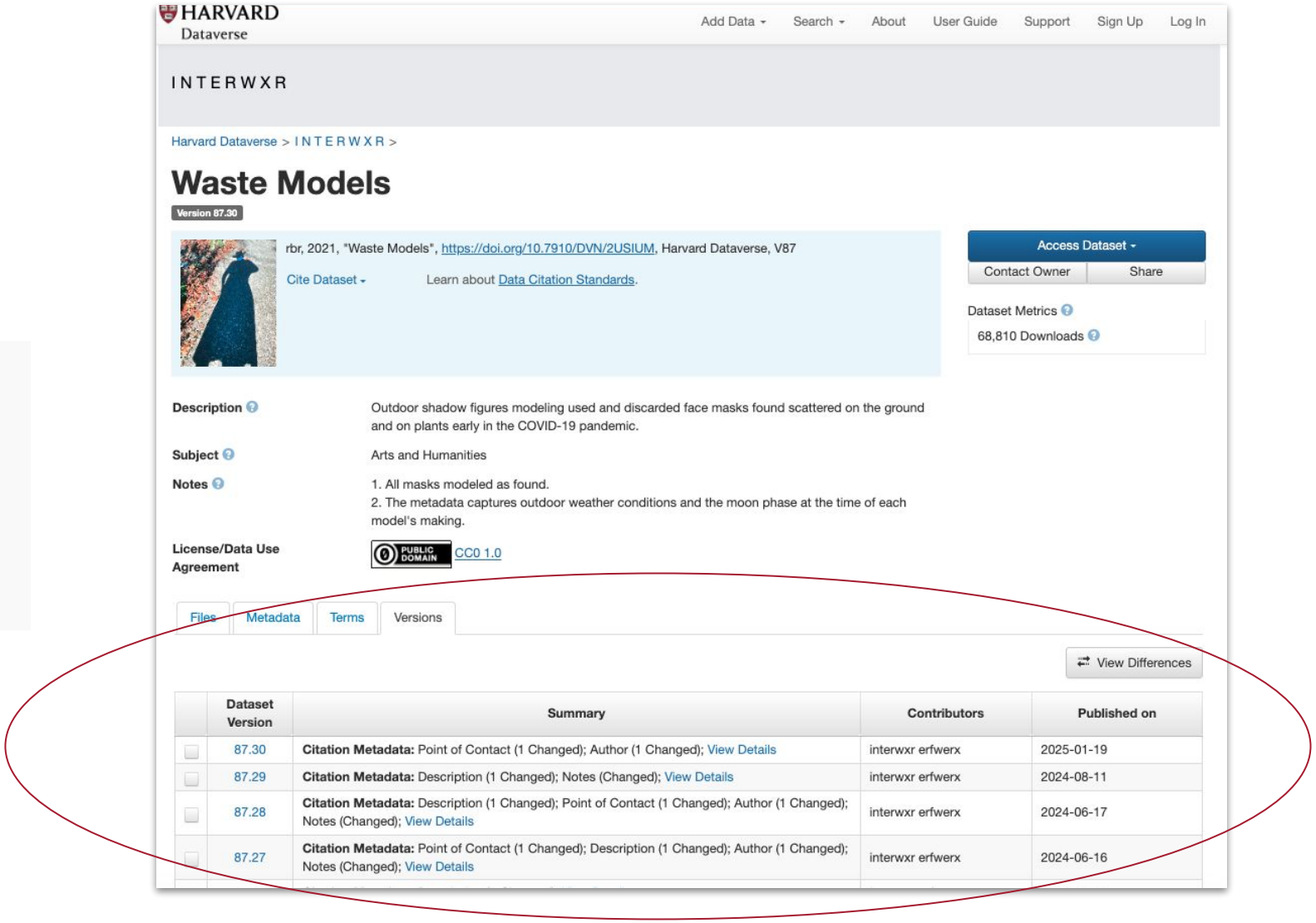
Publish your dataset, receive a citation

Digital Object Identifier (DOI)

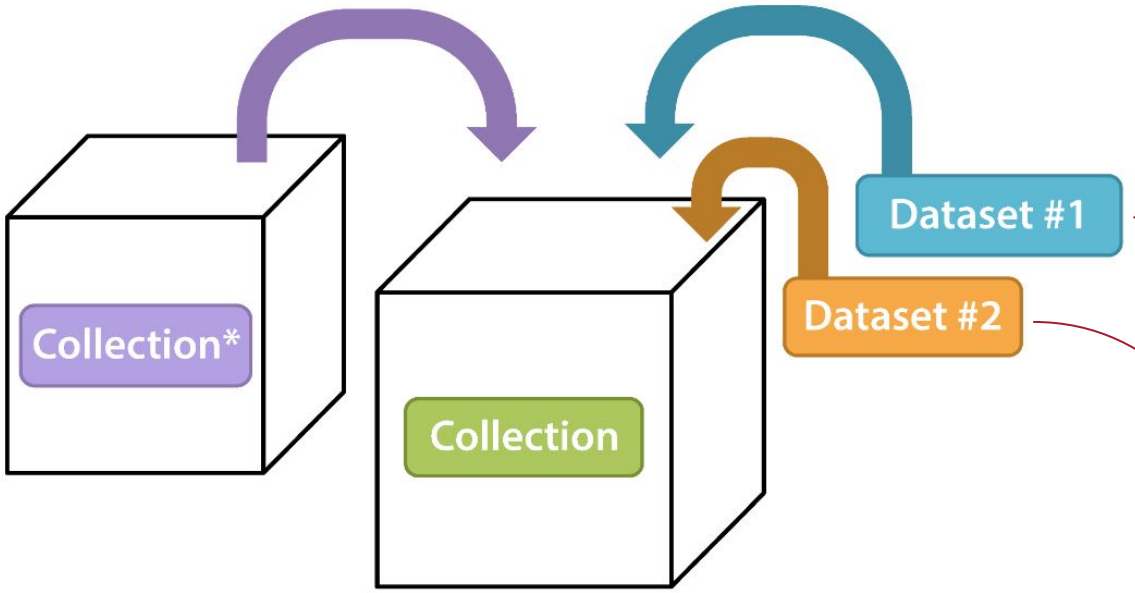


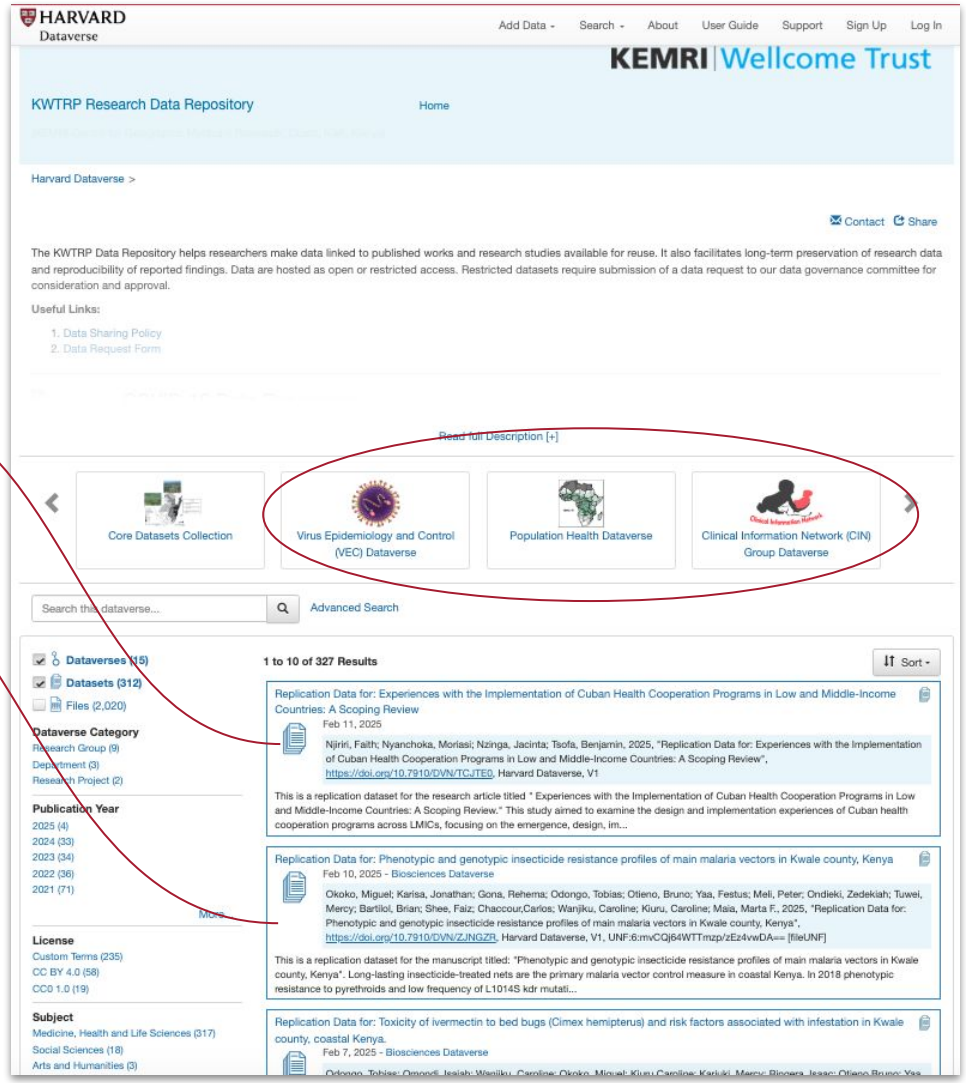
Create and share versions of your data

- Minor version: metadata updates
- Major version: new files, replaced files, etc.



Create a collection for your datasets

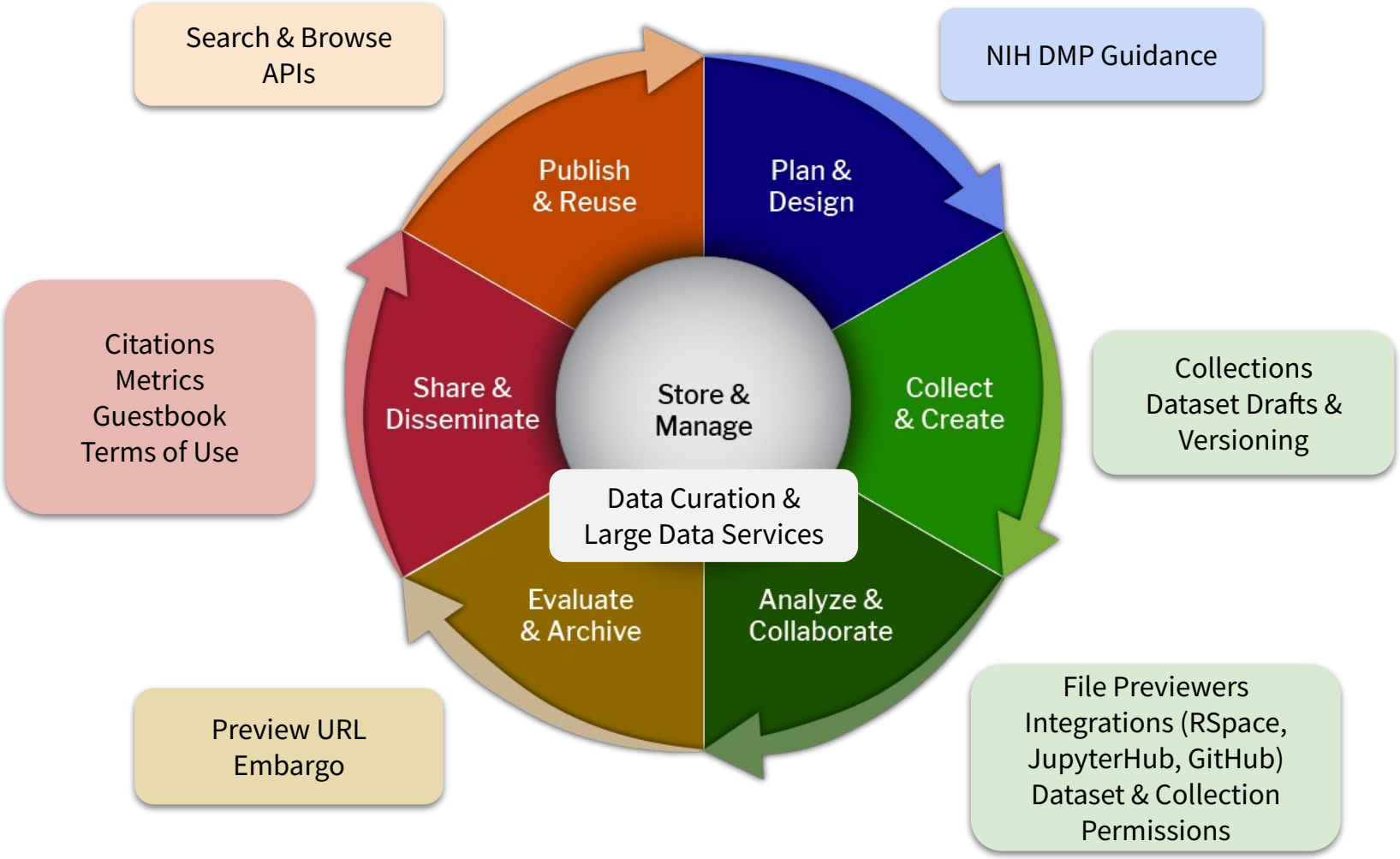




Harvard Dataverse throughout the research lifecycle



Harvard Dataverse features and services



Plan & Design

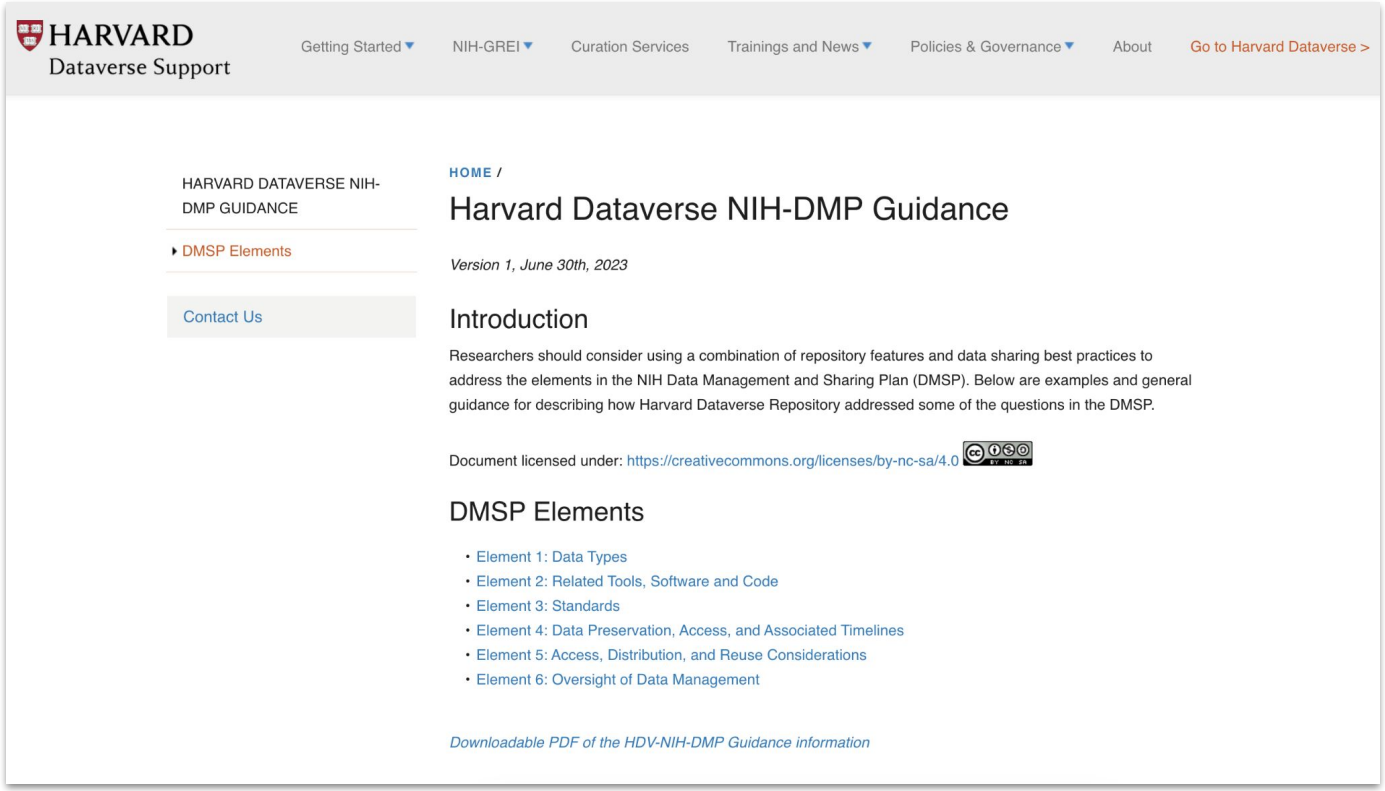
NIH DMP Guidance

NIH Data Management & Sharing Plan

Includes:

- Written plan (2 pgs)
- “maximize the appropriate sharing of scientific data”
- Ideally share in established repositories
- Share as soon as possible

Details: [NIH Scientific Data Sharing Website]



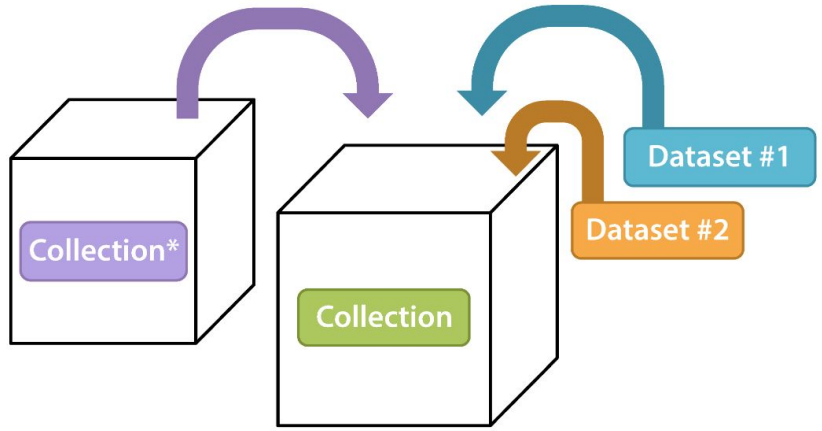
https://support.dataverse.harvard.edu/harvard-dataverse-nih-dmp-guidance

Collect & Create

Collections, drafts & versions

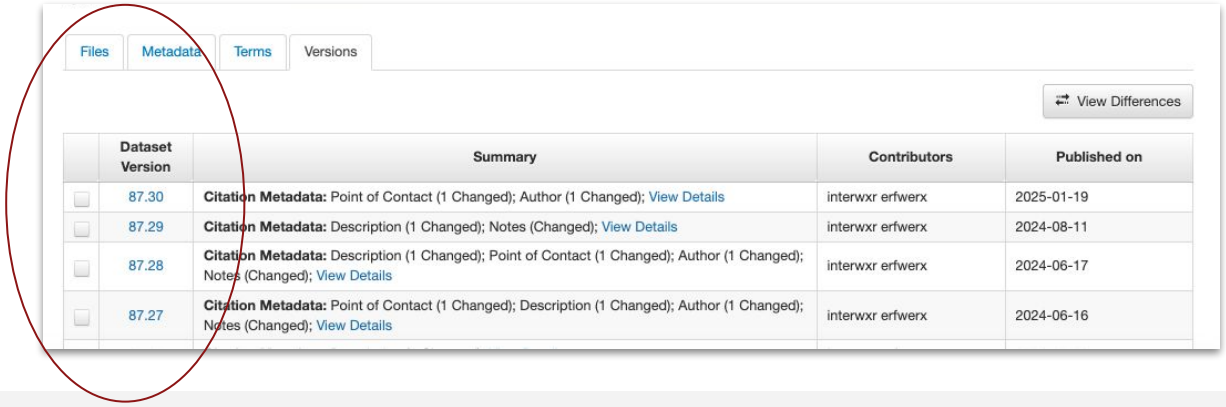
Dataset Collections

- Create containers for your datasets
- Set roles & permissions to allow others to view & deposit datasets to your lab, department, or organization collection



Dataset Drafts & Versions

- Edit your dataset metadata and files while you collect & process your data
- Publish when ready to share publicly
- Add new data and documentation as your project continues or during a new phase



Previewers, tools, integrations & roles

Analyze & Collaborate

Previewers & External Tools

- Tabular files (e.g., STATA, Excel)
- Geospatial files (e.g., NetCDF)
- 3D viewers
- Preview Zip files
- View images, maps, videos
- TurboCurator
- Ask the Data
- …And more

Details: File Previewers & External Tools

Integrations

- GitHub
- Dropbox
- Open Science Framework (OSF)
- RSpace
- JupyterHub
- Whole Tale
- …And more

List: Integrations

Roles & Permissions

Assign collaborators, groups, and others different roles with permissions

- Admin
- Contributor
- Curator
- File downloader
- Member

Details: Roles & Permissions

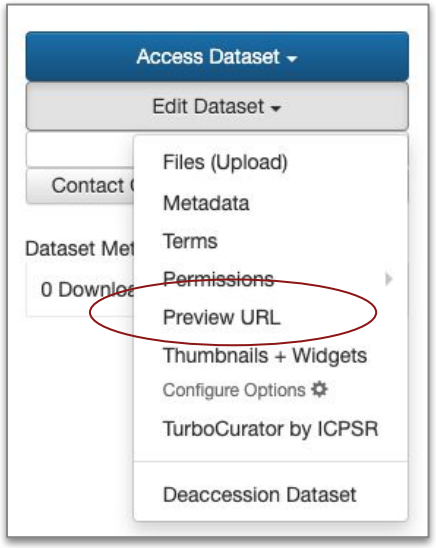
Evaluate & Archive

Preview URL & Embargo

Preview URL

Share link to an unpublished dataset draft, for review

- Share preliminary findings with others
- Peer review

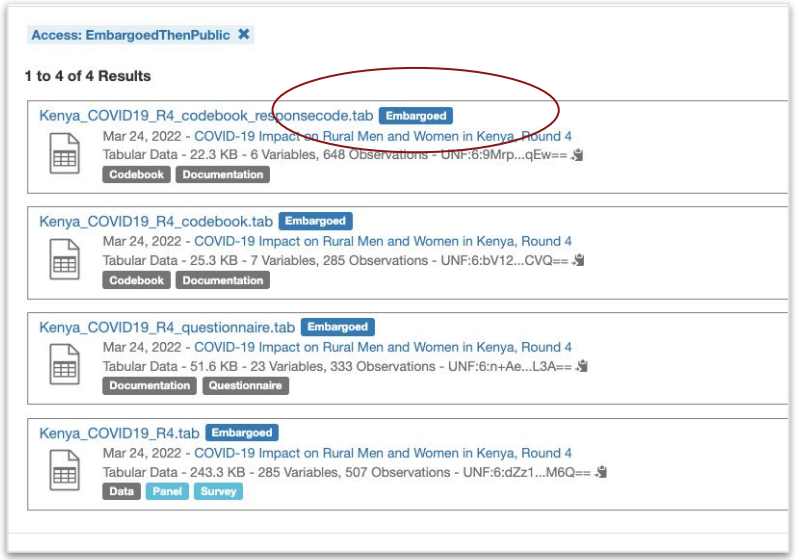


Preview URL Guide: Link

File-Level Embargo

Embargo a data file for a finite period of time

- Meet publication requirements



Embargo Guide: Link

Share & Disseminate

Citations, guestbooks & terms of use

Citations & Metrics

Get a dataset citation (EndNote XML, Ris, BibTeX)

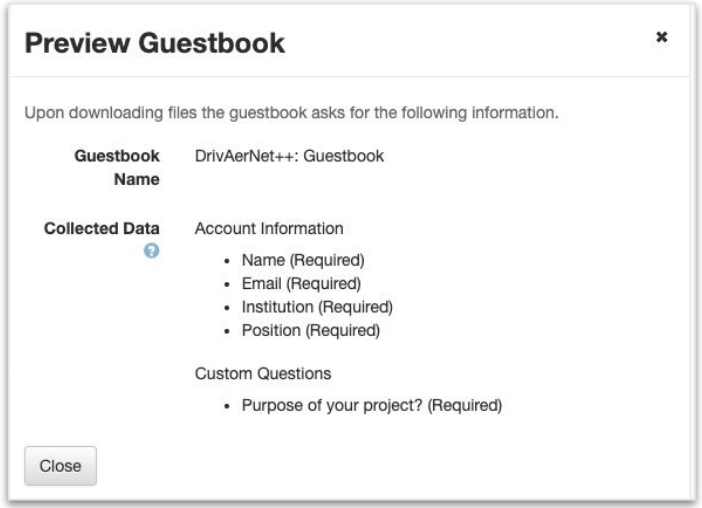
Get dataset download stats



Guestbooks

Use with restricted files

Create a guestbook to collect data about who's using your datasets



Gestbook Details: Link

Terms of use

Apply simple, multiple, and custom licenses and terms of use to your files

Creative Commons, MIT, Apache, others, custom







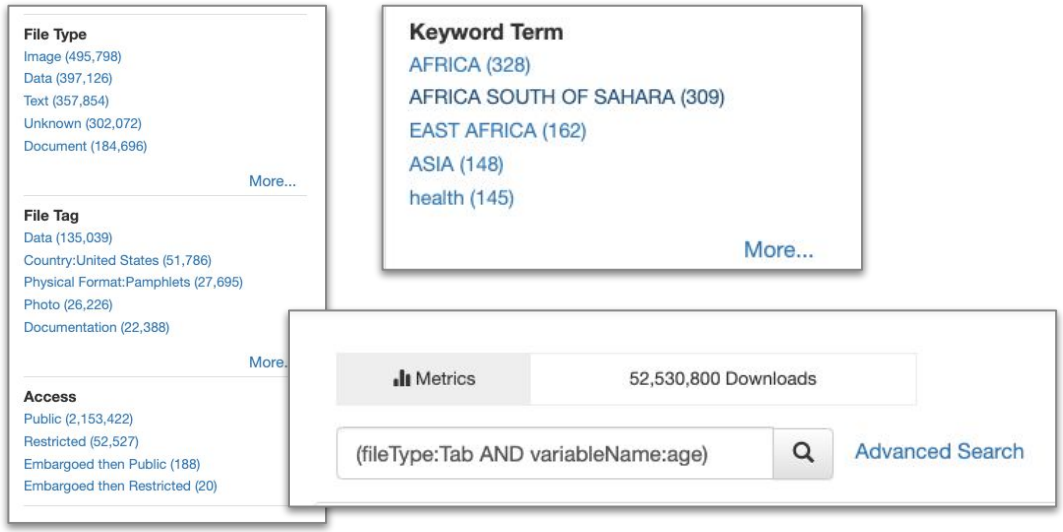
Terms of Use Details: Link

Search, browse & APIs

Publish & Reuse

Search & Browse

- Keyword search
- Advanced search
- Funding agency
- Faceted search
- Sorting
- Search datasets from other repositories



APIs

Programmatic access to Harvard Dataverse using various APIs

- Native (Ex. create datasets, collections)
- Search (Ex. searching, faceting, sorting)
- Metrics (Ex. downloads, datasets created, files uploaded)
- SWORD (RESTful API to deposit files and metadata: Python, Java, R, Ruby, PHP)
- Data Access (download access to files)
- Python library: pyDataverse

Note: No API access for Large Data resources

Visit the API Guide: Link

Store & Manage

Curation & large data services

Curation Services

Range of curation services, including large data

- Free and extended consultations
- Setup and maintain collections
- Upload and maintain datasets

Service Details: Link

Large Data Services

Support for large datasets (over 2.5T or millions of files)

- NESE Globus Tape
- NESE Globus & S3 Disk
- AWS S3

Minimum 6 years storage

Service Details: Link

Trainings & Office Hours: Calendar
Support: support@dataverse.harvard.edu

Harvard Dataverse + AI = Ask the Data

Tell me what you want to know

What is this data about?

Answer please

This data is about traffic accidents in the US.

STATE	COUNTY	MONTH	DAY	HOUR	MINUTE	VE_FORMS	PERSONS	PEDS	NHS	ROAD_FNC	ROUTE	SP_JUR	HARM_EV	MAN_COLL	REL_JUNC
1	97	1	26	15	30	2	2	0	0	6	4	0	12	2	1
1	101	2	1	23	55	1	1	0	1	2	2	0	42	0	1
1	81	2	2	5	10	1	1	0	0	6	4	0	42	0	1
1	101	1	18	22	14	1	2	1	1	12	2	0	8	0	1
1	101	1	20	13	5	1	1	0	1	11	1	0	34	0	1
1	73	1	31	2	40	2	3	0	1	11	1	0	25	0	1
1	73	1	14	19	0	1	2	0	1	11	1	0	34	0	1
1	73	1	24	19	45	2	3	1	0	6	6	0	8	0	2
1	97	1	23	19	34	2	3	0	0	13	2	0	12		
1	97	1	31	23	33	1	2	0	0	4	3	0	30		

Viewing rows 1 through 10 of 38444

Use natural language to query tabular files

- Go to https://dataverse.harvard.edu
- Search for something of interest
- Click on "Files"
- Click on "Tabular Data" (under File Type)
- Click on a file you find interesting
- Ask the Data will appear under File Tools



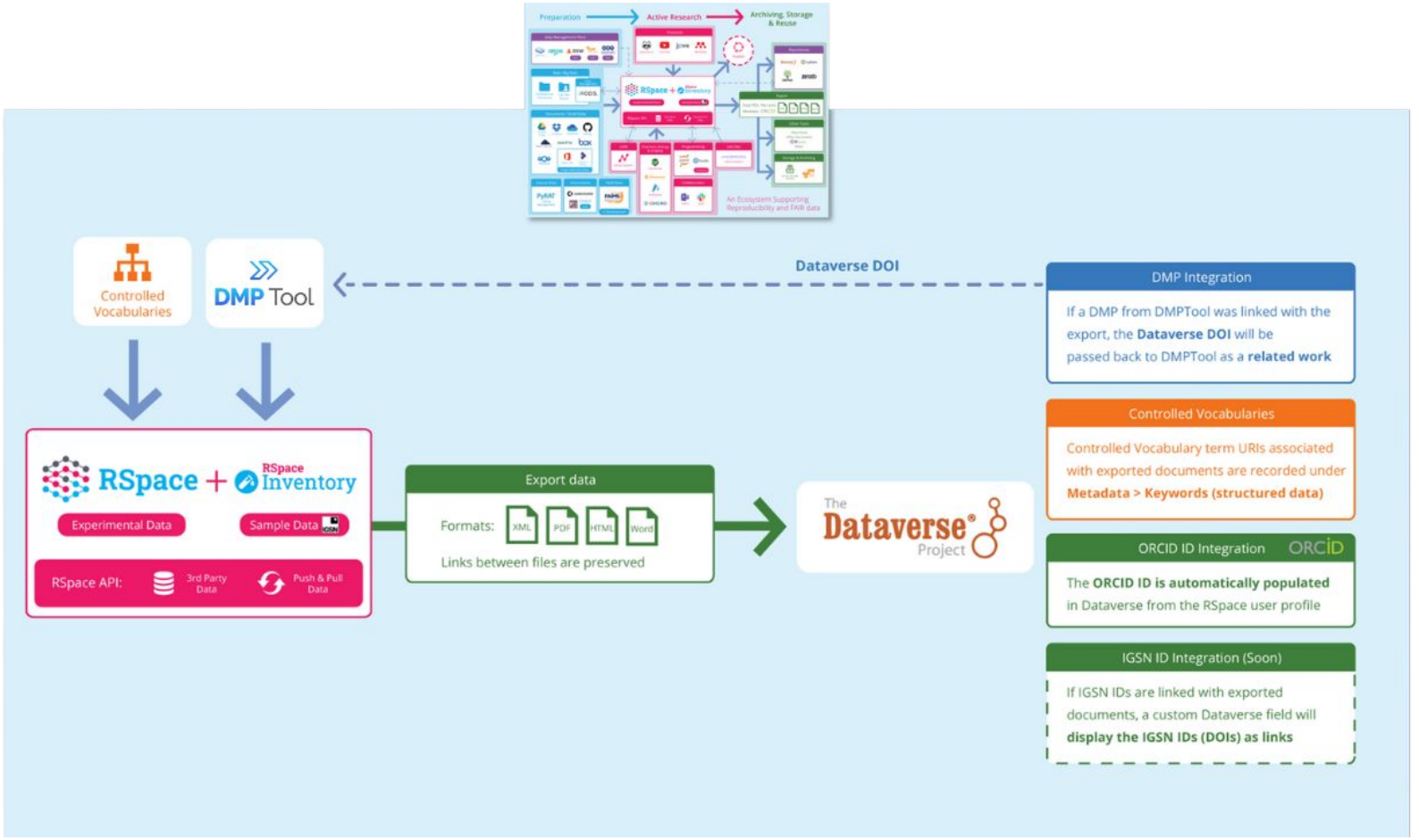


Rory Macneil

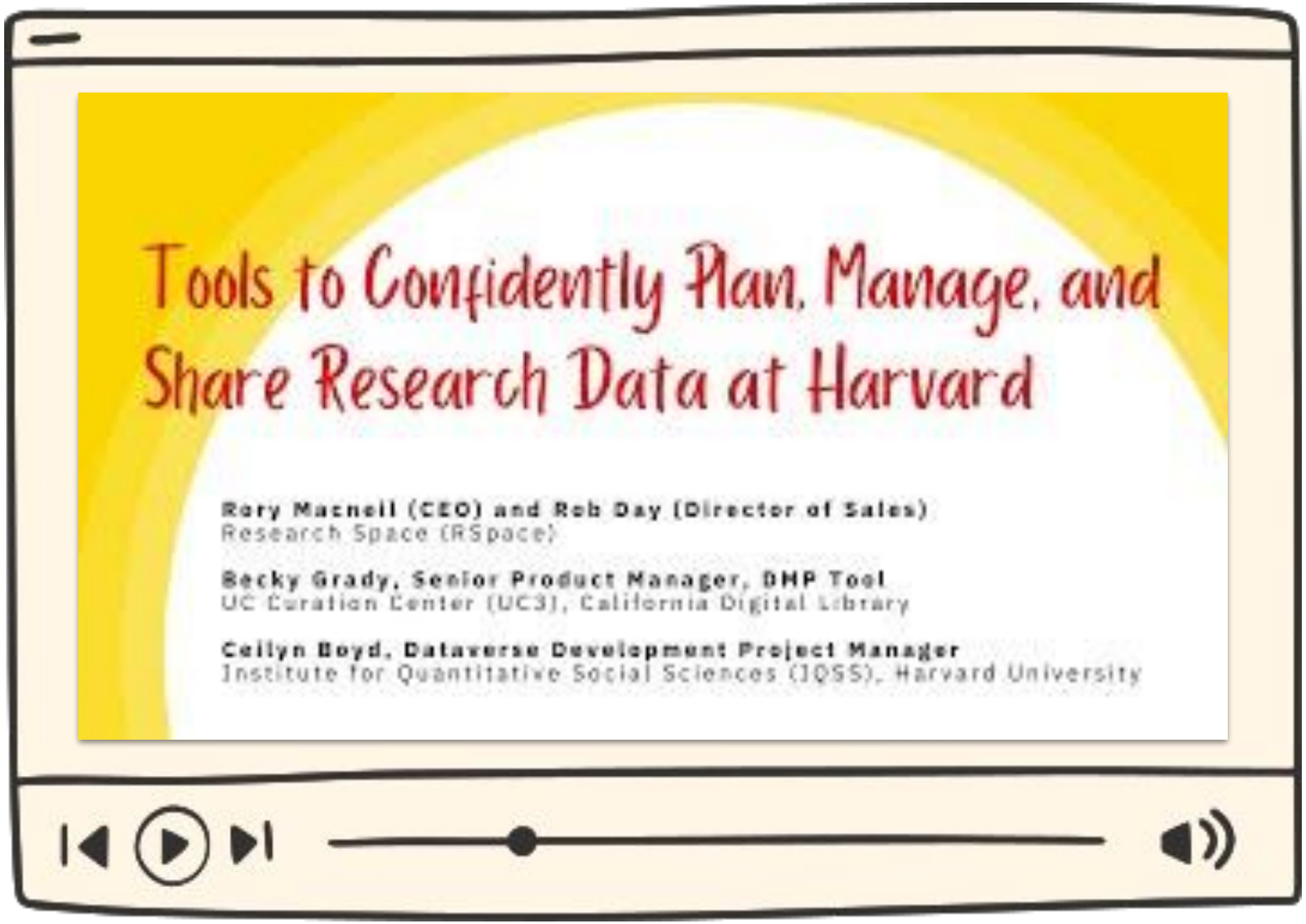
Rob Day

CEO & Director of Sales

Research Space (RSpace)



Find Integration Demo in the YouTube Recording



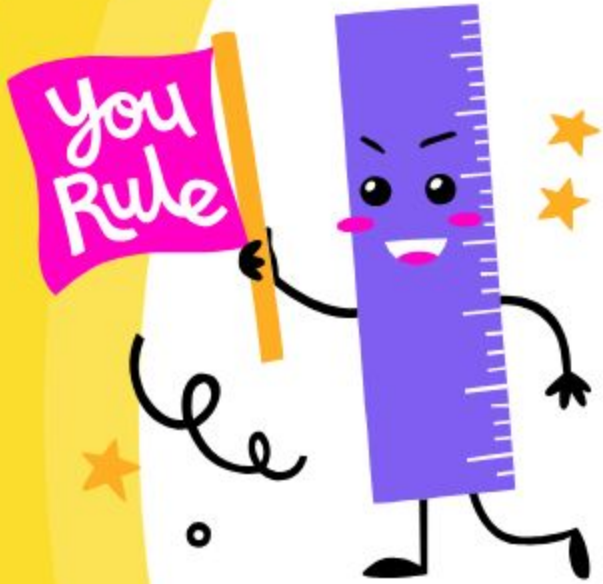
Questions & Discussion

ask a question in the chat

Link Roundup!

- The time is now! Vertical interoperability between research tools: an essential enabler for the FAIRification of data: https://doi.org/10.54900/8m9by-kfy03
- RSpace Example Shared Document: https://demos.researchspace.com/public/publishedView/document/r5OiTJ-rNv69Qd3UdF69cw
- RSpace Tagging & Controlled Vocabularies: https://researchspace.helpdocs.io/article/8ujmvpa1no-tagging-documents
- California Digital Library (CDL) and the Association of Research Libraries (ARL) Machine Actionable Plans (MAP) Pilot: https://bit.ly/mappilot
- DMP Tool User Testing Blog: https://blog.dmptool.org/2024/10/30/behind-the-scenes-insights-from-user-testing-the-new-dmp-tool-designs
- Dataverse Big Data Support: https://guides.dataverse.org/en/latest/developers/big-data-support.html#globus-file-transfer
- Harvard Dataverse Large Data Services: https://support.dataverse.harvard.edu/large-data-support
- Harvard Dataverse Curation and Data Management Services: https://support.dataverse.harvard.edu/curation-services
- Harvard Dataverse Example Dataset with access via Globus: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/PCZYL4
- Harvard Dataverse Example Dataset using tape resources: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/PCZYL4
- Harvard Subscription Data Dataverse: https://dataverse.harvard.edu/dataverse/HarvardSubscriptionData
- RDA DMP Common Standards Working Group: https://www.rd-alliance.org/groups/dmp-common-standards-wg/posts/?post=174771

Happy International Love Data Week 2025



HARVARD LIBRARY

Whose Data is it Anyway?

More events!